Topological observables in lattice QCD and the sampling problem

Second-year PhD seminar · Dottorato in Fisica, Università di Pisa

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θ-dependence: QCD topology and Axion Physics

Gauge fields carry an **integer** topological charge

$$Q = \frac{1}{32\pi^2}\int d^4x\, \varepsilon_{\mu\nu\rho\sigma}\,\mathrm{Tr}\, G_{\mu\nu}G_{\rho\sigma} \;\in \mathbb{Z}$$

and all of θ-physics sits in the free energy

$$F(\theta,T) - F(0,T) = \tfrac{1}{2}\chi(T)\,\theta^2\left[1 + b_2(T)\,\theta^2 + \dots\right]$$

$$\chi = \frac{\langle Q^2\rangle}{V_4}, \qquad b_2 = -\frac{\langle Q^4\rangle - 3\langle Q^2\rangle^2}{12\,\langle Q^2\rangle}$$

Non-perturbative: the lattice is the only first-principles tool.

The QCD axion needs them **at high T**: [1]

Mass
$m_a^2 f_a^2 = \chi(T)$

Relic abundance: misalignment
$\chi(T)$ and $b_2(T)$ over a **range** of temperatures

Thermal production: the source term

$$\Gamma_{\text{top}}(\omega,\vec{p}) = \int dt\, d^3x\, e^{ix_\mu p^\mu}\,\langle q(x)\, q(0)\rangle_T$$

All of them are needed **well above T_c** $\Longrightarrow$ **fine** lattice spacings

The Problem: topological freezing

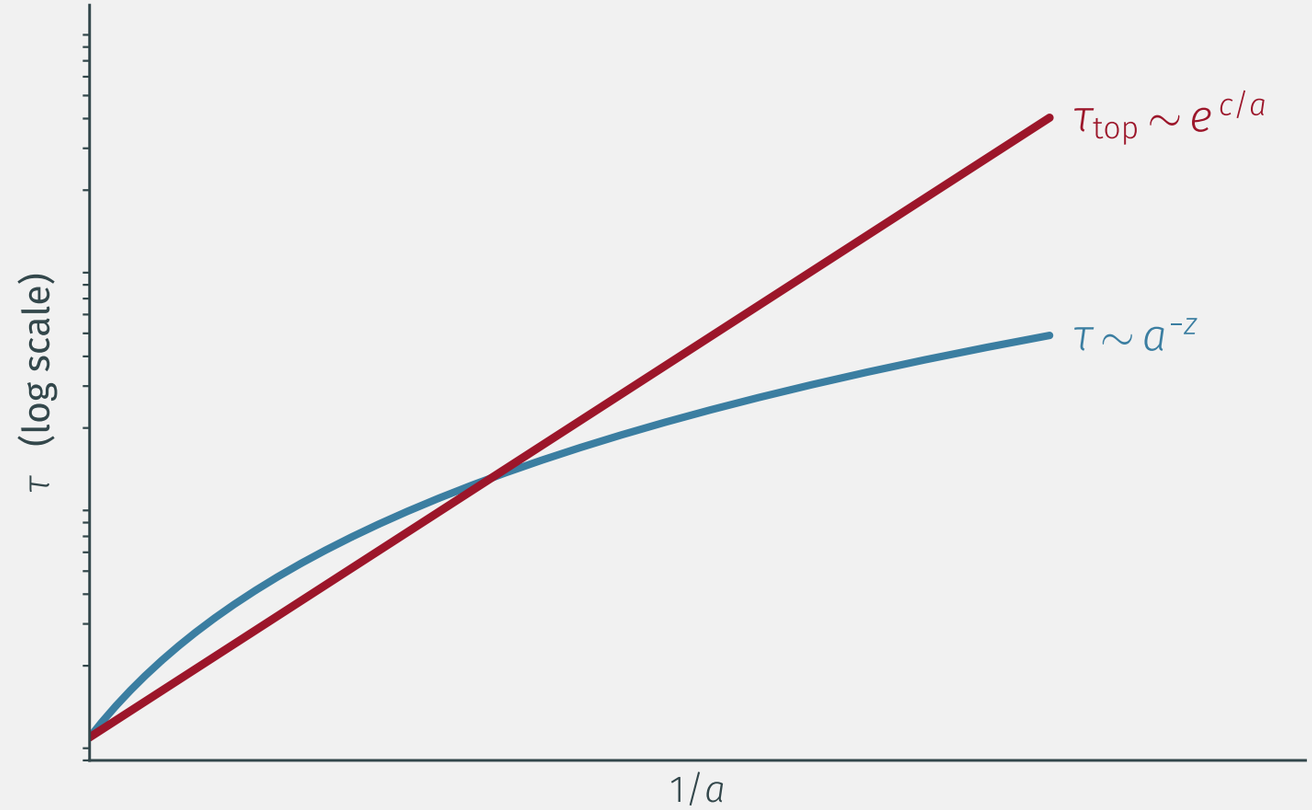
At fixed N_τ, $T = \dfrac{1}{N_\tau\, a}$:
high $T \Longleftrightarrow a \to 0$

Standard critical slowing down

$$\tau_{\mathrm{int}}(\mathcal{O}) \sim a^{-z}$$

Topological freezing [2]

$$\tau_{\mathrm{int}}(Q) \sim e^{\,c/a}$$



On top of that a **physical** suppression: $\chi(T)$ falls steeply, the chain sits at $Q = 0$, and the rare configurations that carry the signal are never sampled.

Two fronts: **what we can measure** and **how we can sample it**

Projects carried in the first two years

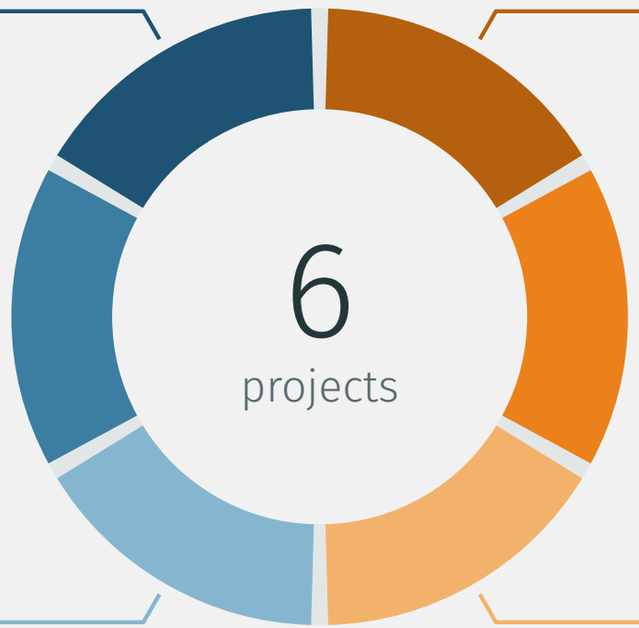
Topological observables
what QCD topology tells us

Algorithmic challenges
how to sample it at all

Topological rate at finite p
real-time transitions · quenched QCD

b_2 across the crossover
θ-dependence · N_f = 2+1 QCD

Lines of constant physics
step scaling + fermion flow

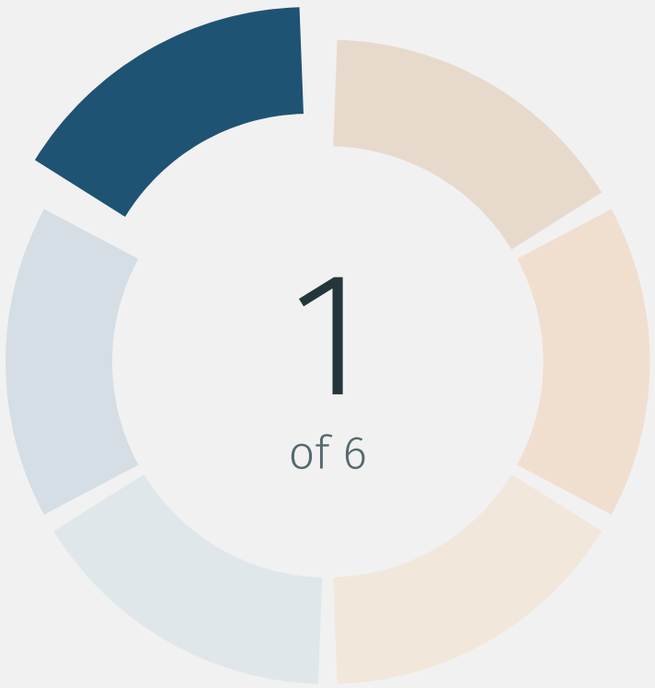


Normalizing flows
learned change of variables · 2D U(1)

Optimal annealing paths
Fisher geometry · CP^{N-1}

Out-of-equilibrium exchange
O-REX · CP^{20}

So **Part II** builds the samplers (hopefully) that **Part I** needs at high T



PART I · TOPOLOGICAL OBSERVABLES

Real-time topological rate at finite momentum

> *How fast does the hot plasma turn topology into axions of momentum $\boldsymbol{p}$?*

arXiv:2608.12066 · quenched SU(3), $T \simeq 1.24\,T_c$ · Lattice 2026

with C. Bonanno, M. D'Elia, G. Gagliardi, A. Giorgieri, F. Sanfilippo, A. Valentino, G. Villadoro

Axion production needs the rate at finite p

Thermal axions obey a **momentum-dependent** Boltzmann equation [3]

$$\frac{df_{\vec{p}}}{dt} = (1 + f_{\vec{p}})\,\Gamma_a^{(+)}(p) - f_{\vec{p}}\,\Gamma_a^{(-)}(p), \qquad \Gamma_a^{(-)}(p) = \frac{\Gamma_{\text{top}}(\omega = E_p,\, p)}{2E_p f_a^2}$$

Higher momenta decouple **later**:
modes may equilibrate at different times $\Rightarrow$ the integrated (sphaleron) rate $\boldsymbol{\Gamma_{\text{top}}(0,0)}$ [4] is not enough.

What the lattice measures

$$G_E(\tau, p) = \int d^3x\, e^{i\vec{p}\cdot\vec{x}} \langle q(\vec{x}, \tau)\, q(\vec{0}, 0)\rangle$$

What we need

$$\Gamma_{\text{top}}(\omega, p) = \frac{2}{1 - e^{-\omega/T}}\,\rho(\omega, p)$$

$$G_E(\tau, p) = -\int_0^\infty \frac{d\omega}{\pi}\,\rho(\omega, p)\,\frac{\cosh\left[\omega(\frac{1}{2T} - \tau)\right]}{\sinh\left(\frac{\omega}{2T}\right)} \quad \Longrightarrow \quad \text{an } \textbf{inverse problem}$$

HLT algorithm

HLT [5]: never reconstruct ρ pointwise, target a **smeared** density

$$\frac{\bar{\rho}(\omega^*, p)}{\omega^*} = -\pi \sum_\tau g_\tau(\omega^*)\, G_E(\tau, p) = \int d\omega\, \Delta_\sigma(\omega, \omega^*)\, \frac{\rho(\omega, p)}{\omega}$$

g_τ minimise

$$F_\lambda[g] = (1-\lambda) \underbrace{A[g]}_{\text{kernel bias}} + \lambda \underbrace{B[g]}_{\text{stat. variance}}$$

Scan λ for the stability plateau; $\sigma/T \in [1.25, 2.5]$, quoted at 1.75.

Three limits, in order

1. continuum $a \to 0$ at fixed R_s
2. smoothing radius $R_s \to 0$ (cooling)
3. smearing width $\sigma \to 0$

$N_\tau = 12, 14, 16$ at $LT = 4 \Rightarrow p/T$ up to 3π

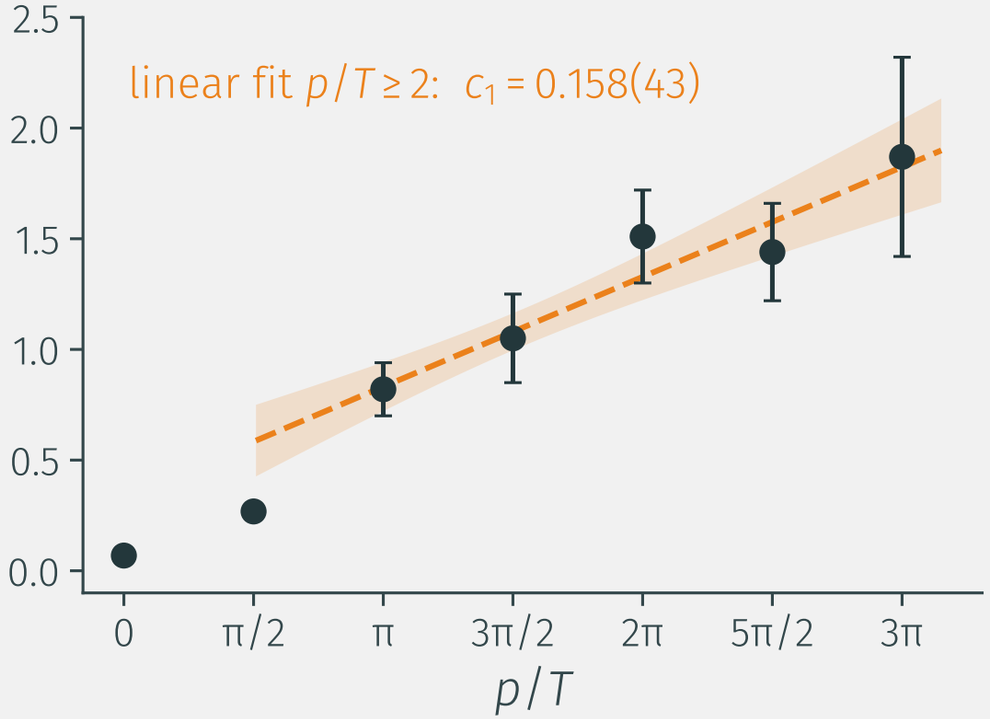
Method 1 · extrapolate, then invert

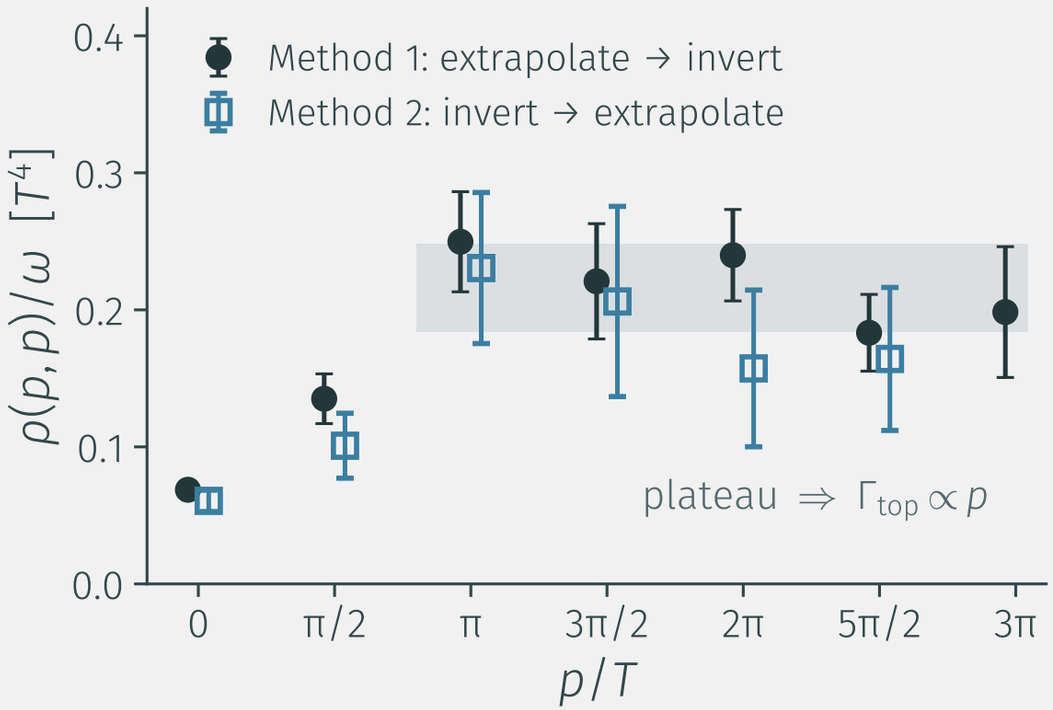
$G_E(\tau, p; a, R_s)$ → $a \to 0$, then $R_s \to 0$ → HLT → $\Gamma_{\text{top}}(\omega, p)$

Method 2 · invert, then extrapolate the rate

$G_E(\tau, p; a, R_s)$ → HLT → $a \to 0$, then $R_s \to 0$ → $\Gamma_{\text{top}}(\omega, p)$

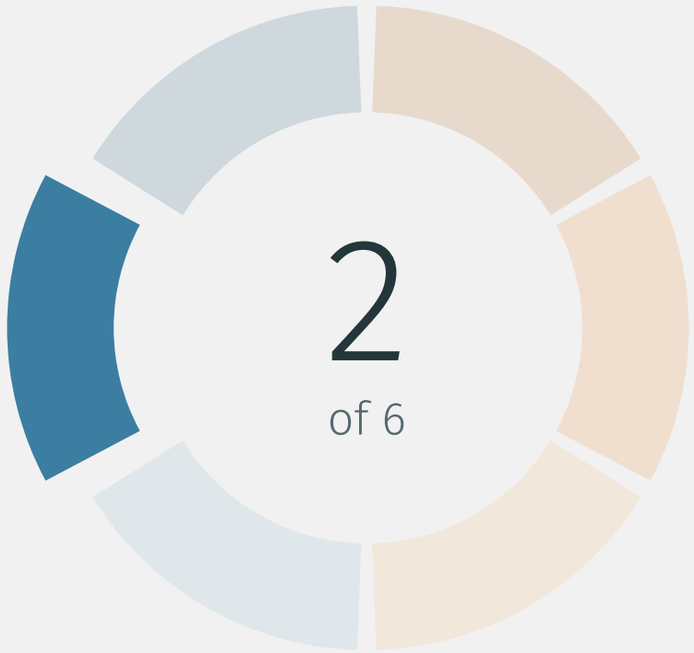
The on-shell rate grows linearly with p





Linear growth for $p/T \geq 2$, as semiclassics predicts [6]
$\Gamma_{\rm top} \propto p\,(\alpha_s T)^3 \left[A \log \frac{pT}{m_D^2} + B\right]$, already at $1.24\,T_c$

The two orderings of the limits **agree** at every momentum.
G_E is *suppressed* with p, but on shell the growth with ω wins.



PART I · TOPOLOGICAL OBSERVABLES

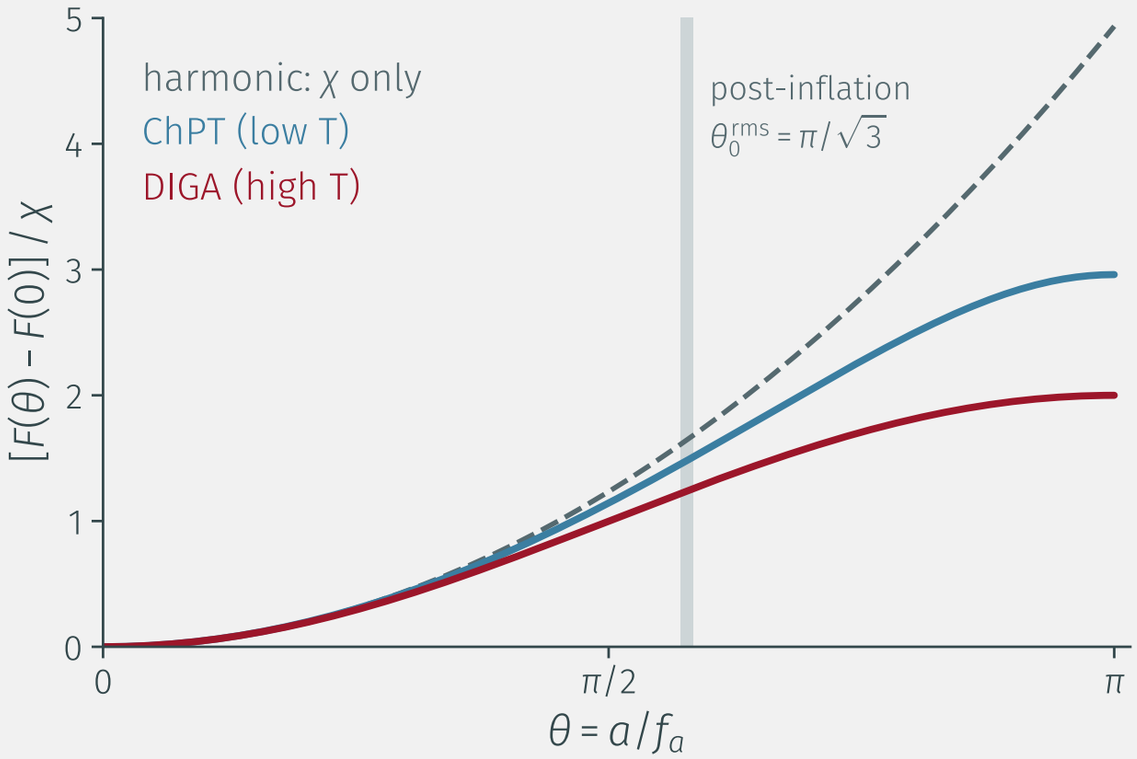
b_2 across the chiral crossover

> *Where does QCD start behaving like a dilute gas of instantons?*

$N_f = 2 + 1$ QCD · $T = 130$ – 200 MeV

Nicolò Francesconi's MSc thesis

Beyond χ: b_2 shapes the axion potential



The axion potential **is** the QCD free energy:

$$V(a) = \tfrac{1}{2}\, m_a^2\, a^2 \left[1 + b_2\, \frac{a^2}{f_a^2} + \dots\right], \qquad m_a^2 f_a^2 = \chi$$

b_2 is the axion self-coupling
$\lambda_a = 12\, b_2\, m_a^2/f_a^2$: at $T = 0$, $b_2 = -0.029$ gives $\lambda_a \simeq -0.35\, m_a^2/f_a^2$, an **attractive** self-interaction [1].

Misalignment does not start at small angles
Post-inflation, $\theta_0^{\rm rms} = \pi/\sqrt{3}$; pre-inflation, the smallest f_a needs $\theta_0 \to \pi$. There the harmonic approximation fails, and the anharmonic terms are set by $b_2(T)$ around $T_{\rm osc}$.

$\chi(T)$ fixes the **curvature** of the potential; $b_2(T)$ fixes its **shape**

From chiral perturbation theory to DIGA

Low temperatures · chiral perturbation theory

$$b_2(0) \simeq -0.029$$

Predicted by the light-quark mass ratios.

High temperatures · dilute instanton gas approximation

$$b_2^{\mathrm{DIGA}} = -\frac{1}{12}, \qquad b_4^{\mathrm{DIGA}} = \frac{1}{360}$$

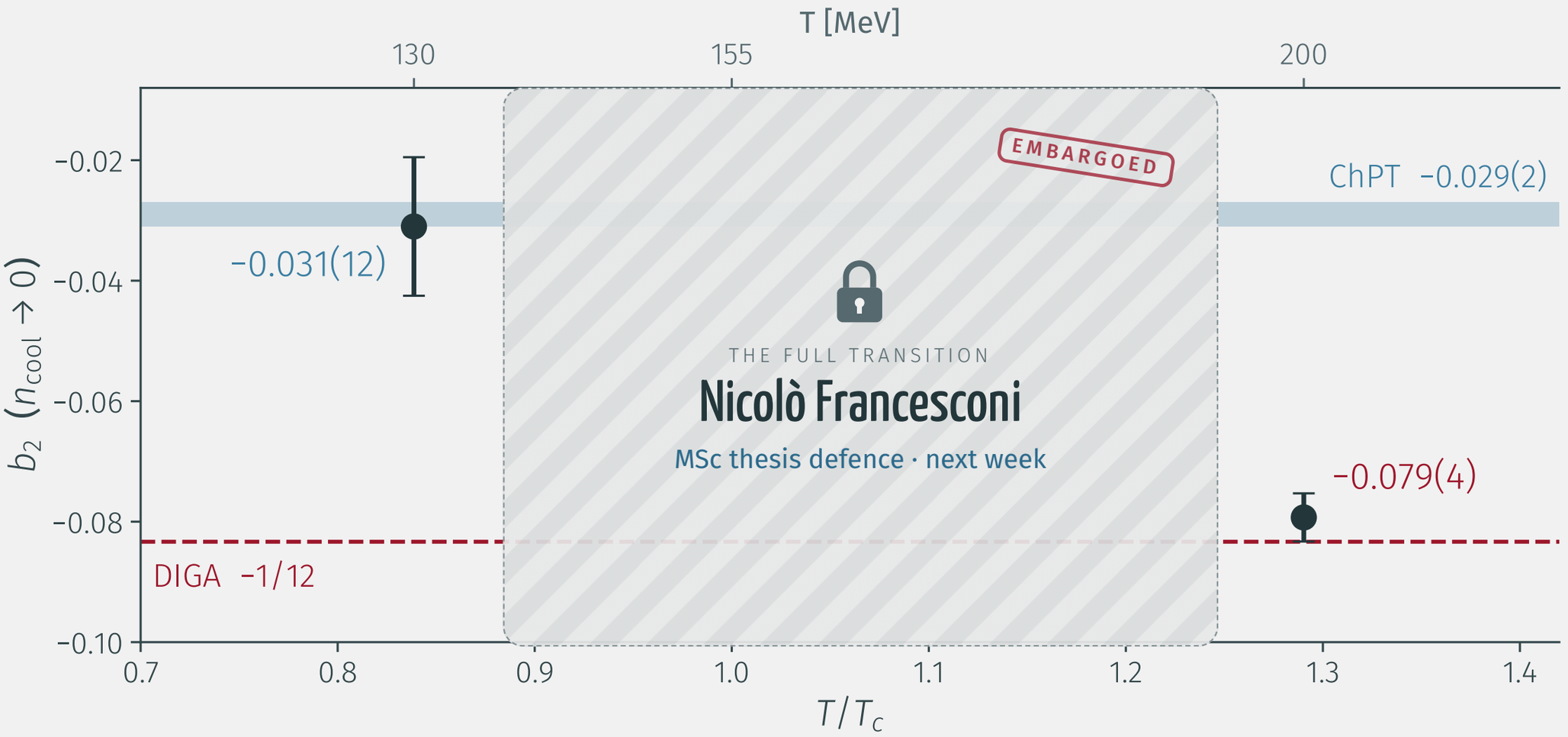
Characteristic values of the DIGA regime.

Pure Yang–Mills vs. QCD

- Pure SU(3) Yang–Mills: first-order deconfinement transition [20].
- Physical QCD: crossover; the approach to the DIGA regime is not fixed by the transition temperature.

The temperature dependence of b_2 provides a **diagnostic of the approach to the DIGA regime**. Establishing the onset of DIGA is important for assessing the reliability of high-temperature extrapolations of $\chi(T)$, a key input for QCD axion cosmology.

b_2 across the crossover in Nf = 2+1 QCD



$T = 130$ MeV: $-0.031(12)$, on the **ChPT** value

$T = 200$ MeV: $-0.079(4)$, on the **DIGA** value

The whole ChPT → DIGA transition, resolved for the first time in $N_f = 2 + 1$ QCD.
How QCD gets from one end to the other: **Nicolò's MSc defence, next week**.



Lines of constant physics at high temperature

> *Higher T means finer a:*
> *charm is dynamical ($N_f = 2 + 1 + 1$), and the $T = 0$ route costs $\propto a^{-4}$.*

gradient-flow step scaling · bare parameters · fermion flow

The gradient flow

Evolve the links $V_\mu(x,t)$, from $V_\mu(x,0) = U_\mu(x)$, in a fictitious **flow time** t [21]:

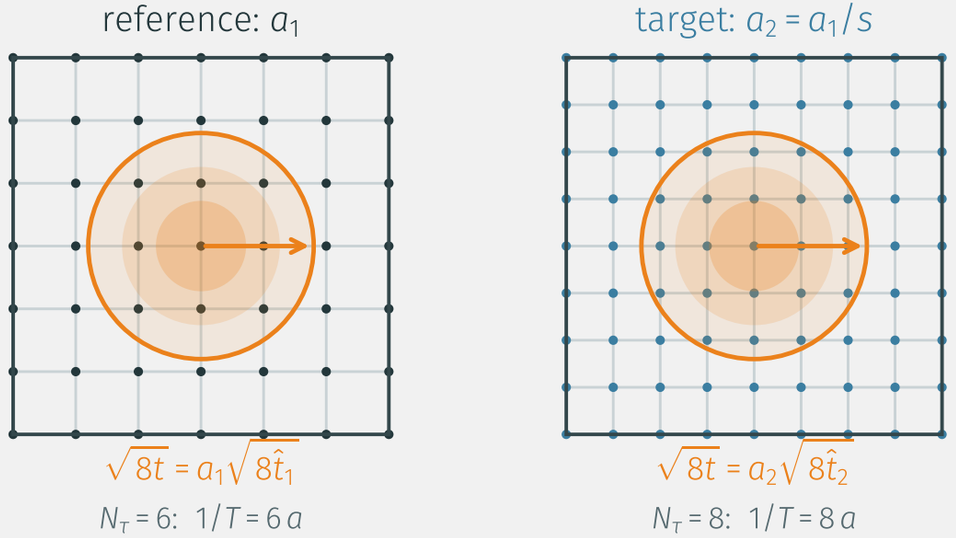
$$\partial_t V_\mu(x,t) = -g_0^2 \left\{\partial_{x,\mu} S_W[V]\right\} V_\mu(x,t)$$

UV fluctuations are averaged over a radius $\sqrt{8t}$; flowed gluonic observables are **finite**, with no further renormalisation:

$$g^2_{\text{flow}}(\mu) = \frac{16\pi^2}{3}\, t^2 \langle E(t)\rangle \Big|_{\mu=1/\sqrt{8t}}$$

The step-scaling procedure

1. reference ensemble (β_1, a_1) on $N_s^3 \times N_\tau$
2. target $(sN_s)^3 \times sN_\tau$ at trial β_2: same box and T **iff** $a_2 = a_1/s$
3. flow both, read $\hat{t}_1, \hat{t}_2$ where g^2_{flow} is equal
4. $s(\beta_2) = \sqrt{\hat{t}_2/\hat{t}_1}$; scan β_2 until s hits its target



One physical flow radius, more sites on the finer lattice:

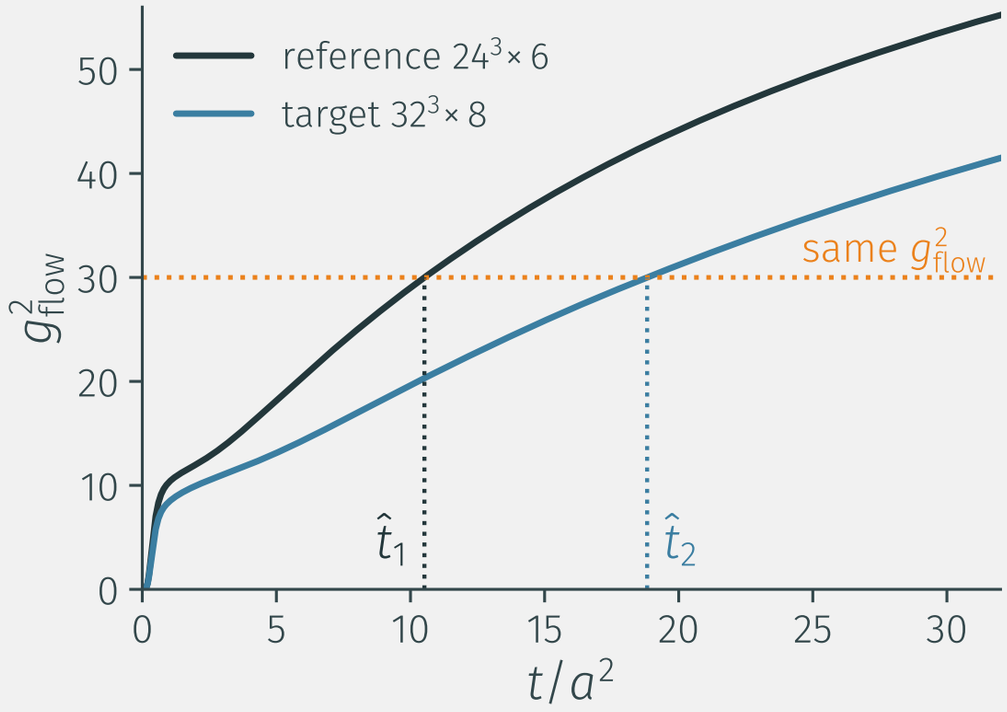
$$\hat{t}_2/\hat{t}_1 = s^2$$

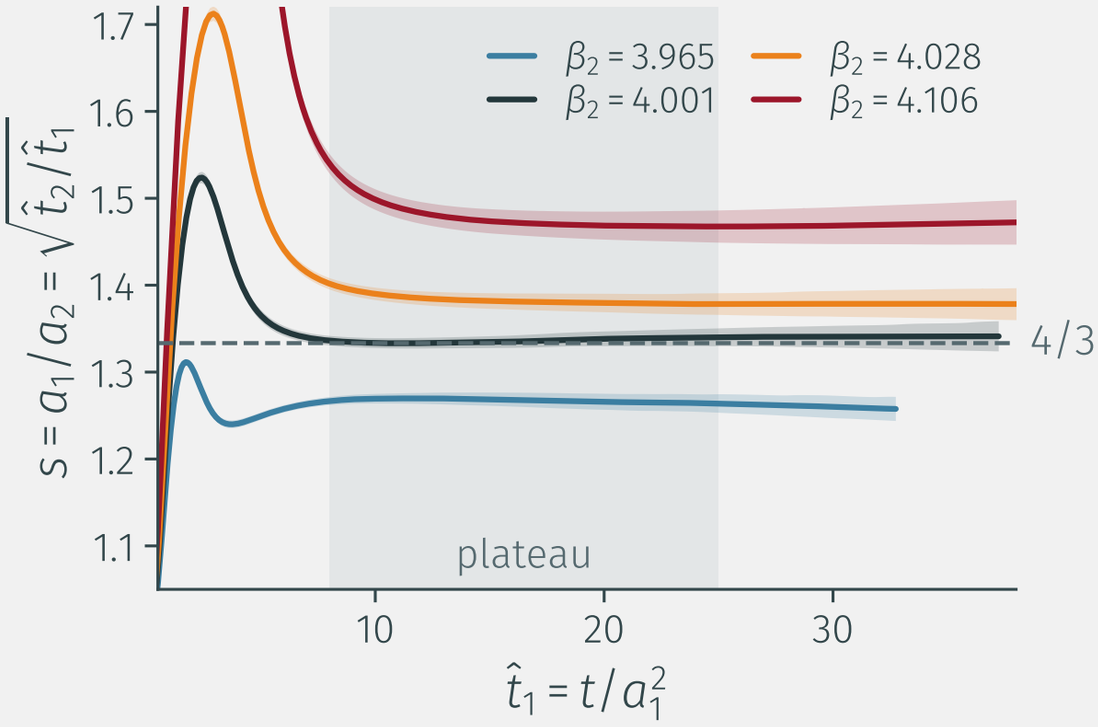
Why it suits high T:

1. Only a local gluonic observable, measured **on the finite-T ensembles**:
2. No hadron spectroscopy
3. No $T = 0$ runs at a^{-4} cost.

Step scaling: the gauge coupling from the flow

Same g^2_{flow} $\Rightarrow$ same physical flow time on both lattices [7]: $t = a_1^2\hat{t}_1 = a_2^2\hat{t}_2 \implies s \equiv \dfrac{a_1}{a_2} = \sqrt{\hat{t}_2/\hat{t}_1}$





Purely gluonic and local: it works **at finite T**.

A flat plateau in $\mathbf{1} \ll \hat{t} \ll N_\tau^2$, moving monotonically with β_2: scan and interpolate to the target s.

theory	step	T [MeV]	$\beta_2^\star$
$2+1$ (check)	$24^3\times 6 \to 32^3\times 8$	498	4.121
$2+1+1$	$24^3\times 6 \to 32^3\times 8$	498	4.004
$2+1+1$	$24^3\times 8 \to 36^3\times 12$	374	4.040

Agrees with the BMW scale setting [8] at the **per-mille** level

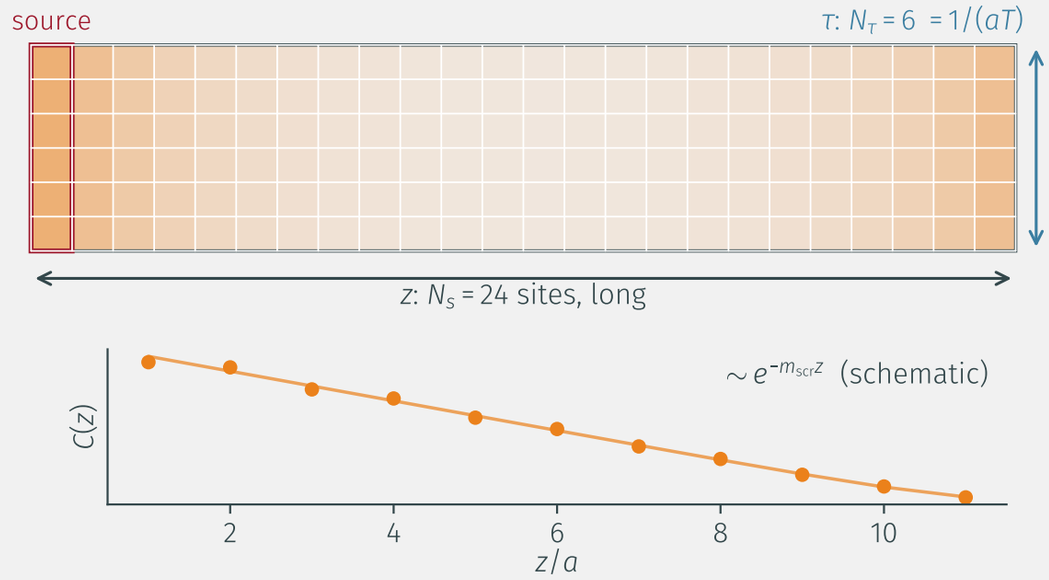
Fixing the quark masses: screening masses

Why not the time direction?
At finite T it is only $1/T = N_\tau a$ long: $N_\tau/2 = 3$–4 independent points, far too few to isolate a ground state.

Use a long spatial direction
$L = N_s a$ with $LT = 4$:

$$C(z) = \sum_{x,y,\tau} \langle O(x,y,z,\tau)\, O^\dagger(0) \rangle \sim e^{-m_{\mathrm{scr}} z}$$

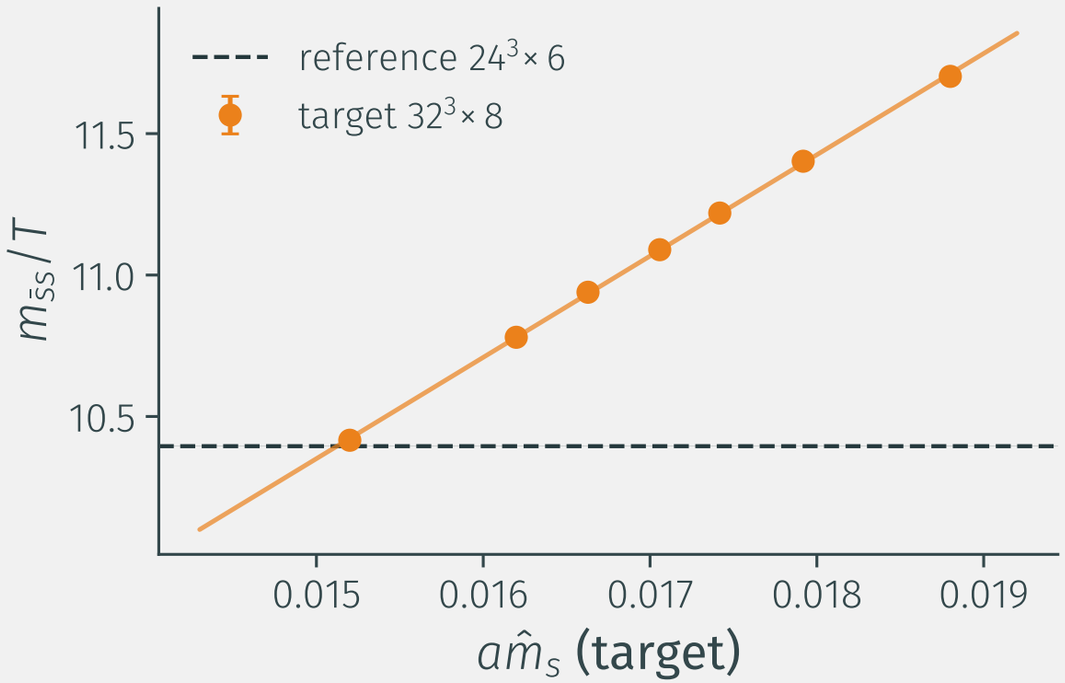
m_{scr} is an eigenvalue of the transfer matrix along z: the inverse correlation length of the plasma in that channel.
Not a pole mass: the bath breaks Lorentz symmetry.

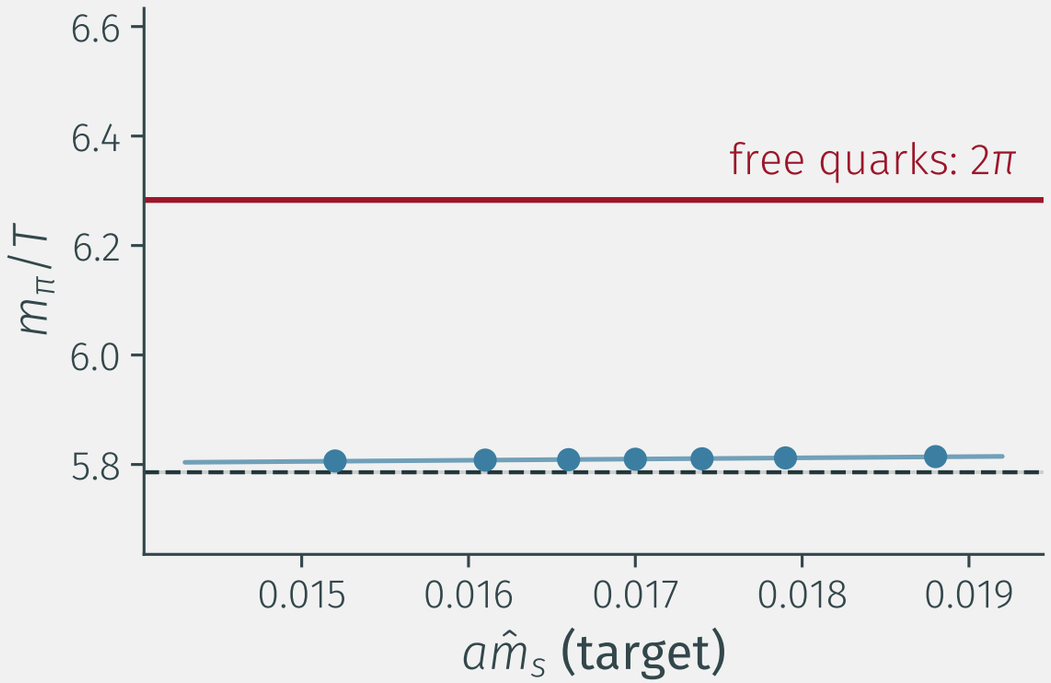


The matching

- Same T on both lattices, so equal m_{scr}/T means equal physics.
- Scan $a\hat{m}_s$ on the target (ratios m_s/m_l, m_c/m_s fixed) and interpolate to the reference value.
- Staggered fits: $C(z) = A\cosh[m(z - \frac{L}{2})] + (-1)^z A' \cosh[m'(z - \frac{L}{2})]$.

Quark masses: screening masses not reliable at high T





At high T every quark carries at least πT
$m_{\text{scr}} \to 2\pi T + \mathcal{O}(m^2/T)$ [9]: the dependence on the bare mass is washed out.

Already at $T \simeq 500$ MeV
$\bar{s}s$ still tunes; the pion sits **7–8 % below 2π**.
Towards 1 GeV even s and c saturate.

We need a **short-distance** fermionic observable, decoupled from the thermal bath

Flowed quark bilinears

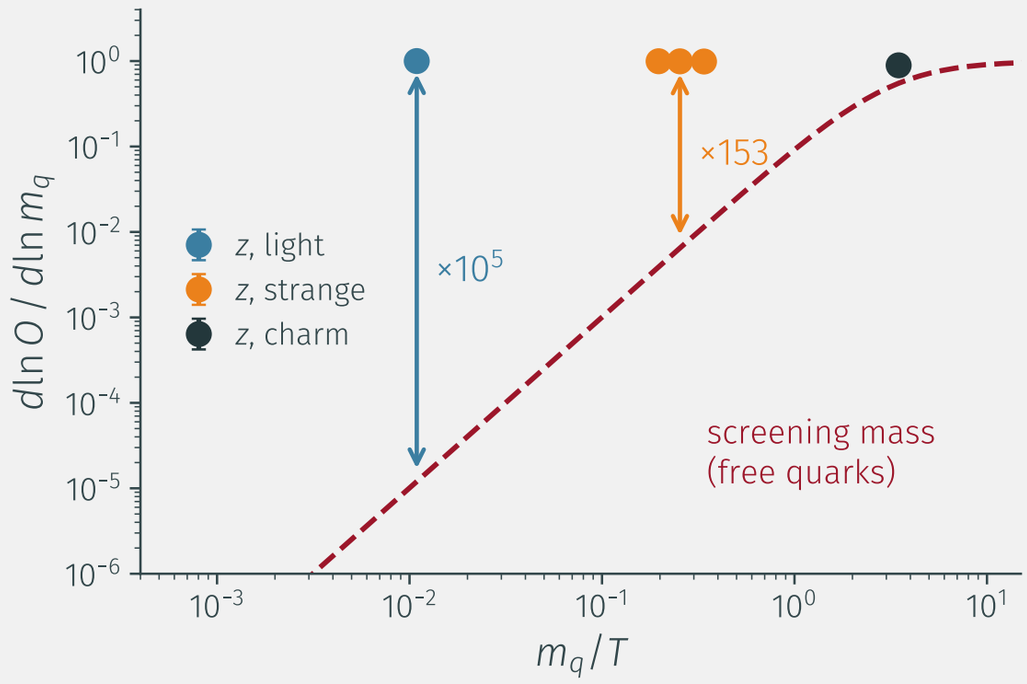
WORK IN PROGRESS

Flow the **quark fields** too (Lüscher [10]) to $\sqrt{8t} = c\,L$ and take

$$z_f = \mathcal{N}_m^{-1}\,\frac{L}{t}\,\frac{\langle \bar{\chi}_f \chi_f \rangle_t}{\langle \bar{\chi}_f \gamma_\mu \overleftrightarrow{D}_\mu \chi_f \rangle_t} \;=\; L\, m_f^{\mathrm{R}}(\mu \sim 1/L)$$

- field renormalisation **cancels** in the ratio
- numerator chirality-odd (needs m), denominator even
- staggered: a **two-link** flow keeps the remnant $U(1)_\varepsilon \Rightarrow$ no additive renormalisation, **one** Z for all flavours

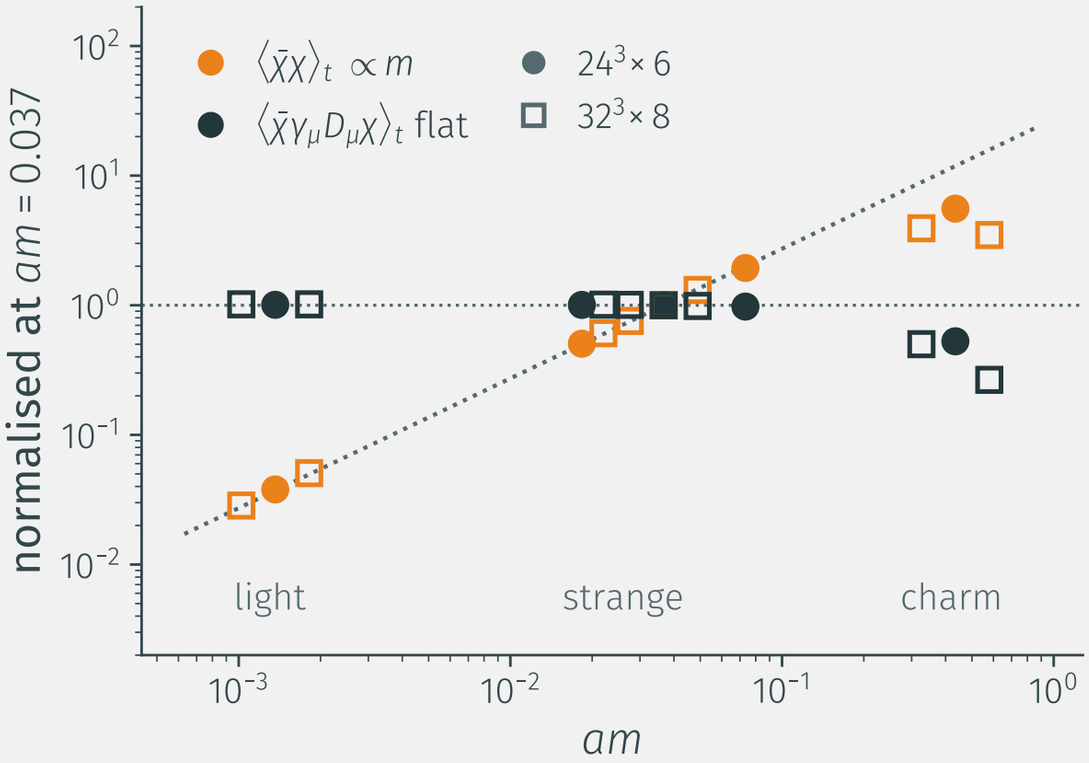
New `fermion_flow_meas` implemented in OpenStaPLE (GPU, OpenACC).



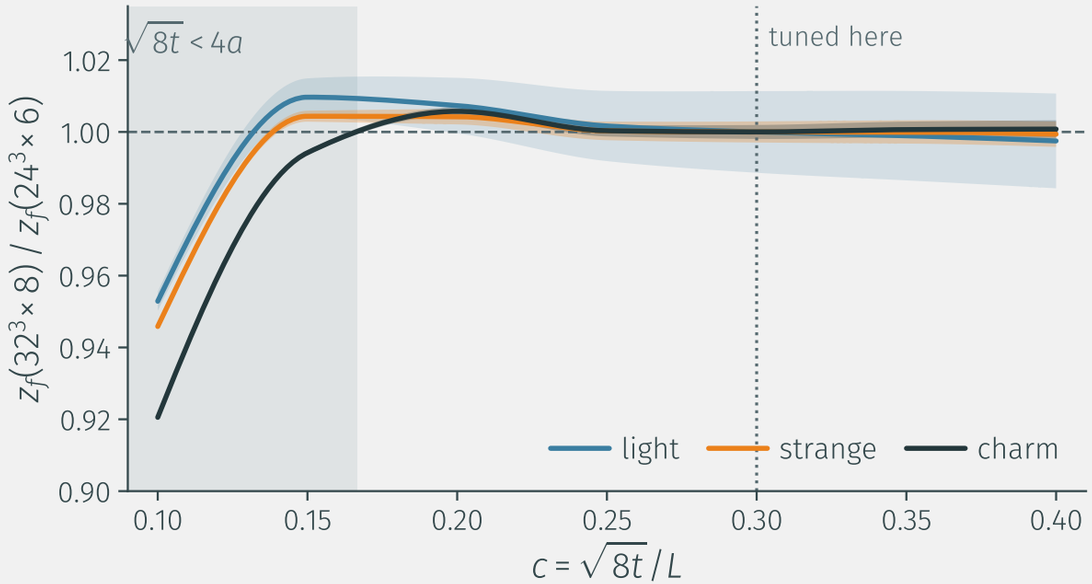
Better Sensitivity than Screening masses: 1% of m moves z by 1%

z is a genuine renormalised mass

WORK IN PROGRESS

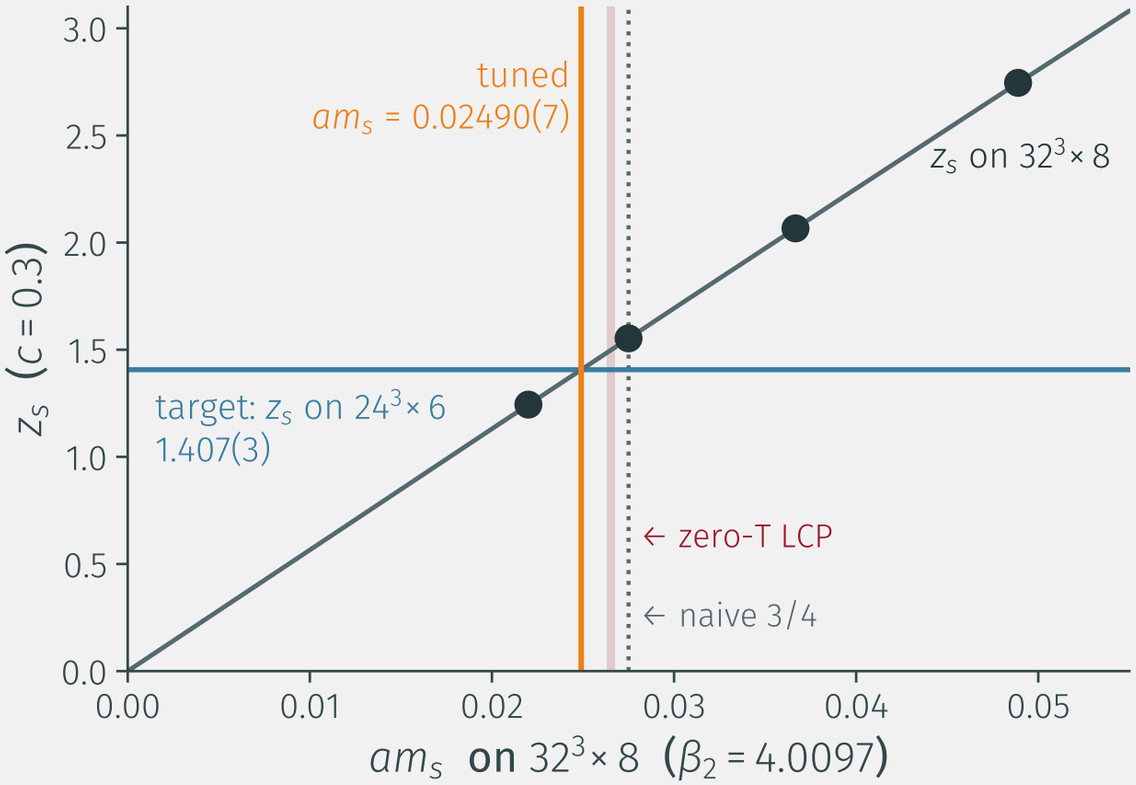


Numerator $\propto m$ for a large range of masses, denominator flat.
$z(m \to 0) = 0.0001(1)$ on $32^3 \times 8$: **no additive renormalisation**



Masses tuned at $c = 0.3$ only: the whole $z(c)$ agrees between the two spacings within 0.5% once $\sqrt{8t} \gtrsim 4a$. **The flow removes the cutoff.**

First tuning: β = 3.873 → 4.010 WORK IN PROGRESS



	$am^{\text{tuned}}(\beta_2)$	$am(\beta_2)/am(\beta_1)$
light	0.000934(11)	0.685(8)
strange	0.02490(7)	0.679(2)
charm	0.2973(6)	0.684(1)

- **0.3 %** on m_s in **~20 A100-hours**, on existing configurations
- the **same** factor for l, s, c, flat in c
- flavour ratios reproduced:
  $m_c/m_s = 11.94(7)$ (LCP 11.85), $m_s/m_l = 26.7(2)$ (26.91)

Open "Problem":
The overall scale sits **6–7 % below** the zero-T LCP, far beyond the errors:

- cutoff effects constant in c? running?
- A third lattice spacing will settle this.



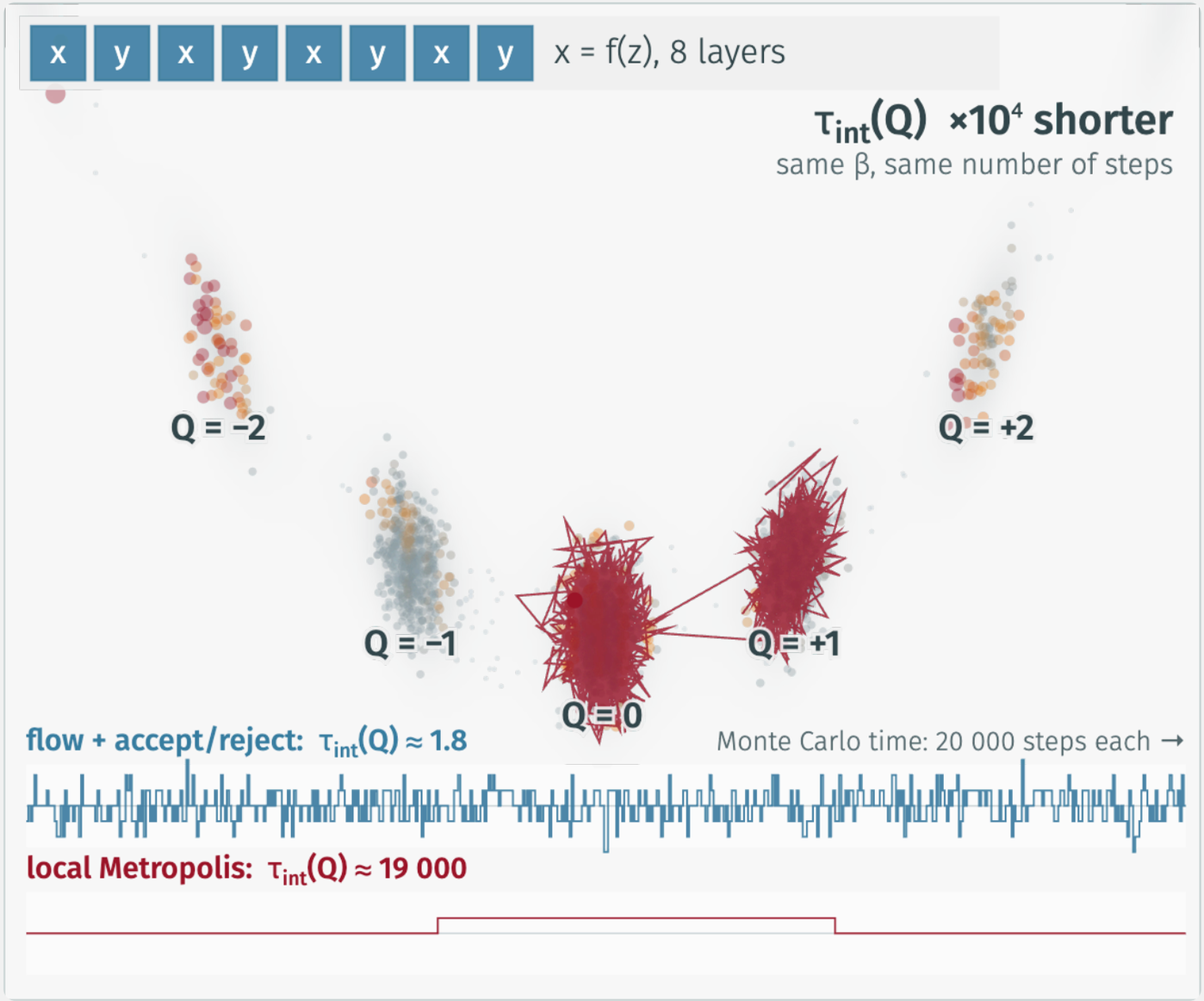
PART II · ALGORITHMIC CHALLENGES

Sampling topology with normalizing flows

> *Can a learned map jump over the barrier, instead of climbing it?*

2D $U(1)$ gauge theory · exactly solvable testbed · vs HMC

A flow is a learned change of variables



Toy Model $p_\beta \propto e^{-S_\beta}$:

- five **sectors** Q behind barriers $\propto \beta$.
- Draw $z \sim \mathcal{N}(0, \mathbb{1})$
- Push it through $x = f_\theta(z)$:

$$q(x) = q_0(z) \left| \det \partial f_\theta / \partial z \right|^{-1}$$

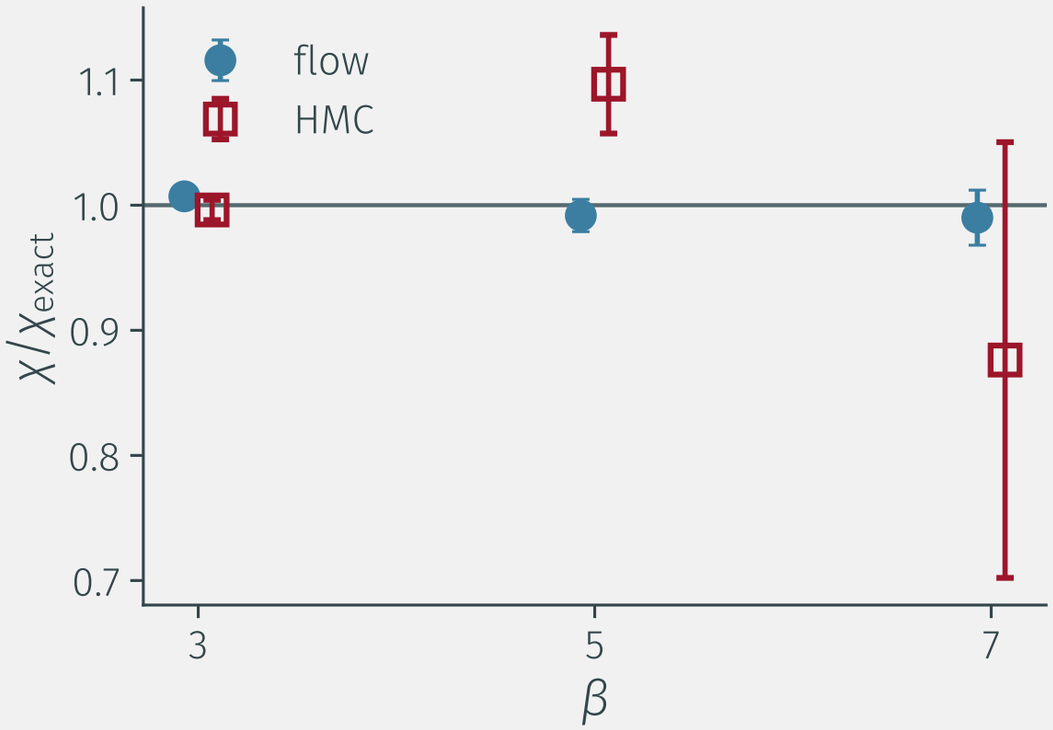
Learned from the action alone
8 spline coupling layers, each moving **one** coordinate given the other (cheap Jacobian). Trained in **5 s** by minimising $\mathbf{KL}(q\|p)$ with β annealed from 0 to 7.

Never trusted
$w = p/q$ is known for every sample: ESS $= 84\%$. The flow **under-fills** $Q = \pm 2$ by up to $2\times$ (hollow bars); reweighting or Metropolis restores the exact $P(Q)$.

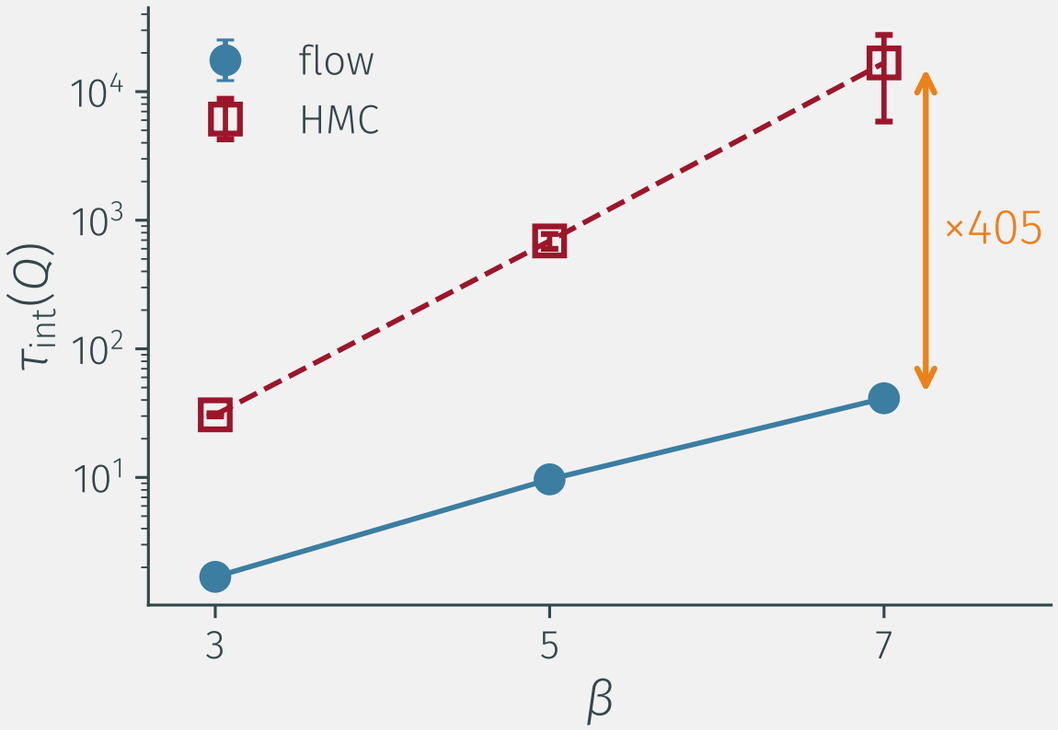
Independent proposals beat freezing
Flow beats Local Metropolis by **4 orders of magnitude** in $\tau_{\mathrm{int}}(Q)$ at $\beta = 7$.

Flow vs HMC

Testbed: 2D $U(1)$, $\pi_1(U(1)) = \mathbb{Z}$, freezes like QCD and is **exactly solvable**;
the flow reproduces the free energy to 5 digits ($-F = 568.424(48)$ vs 568.422 at $\beta = 7$).

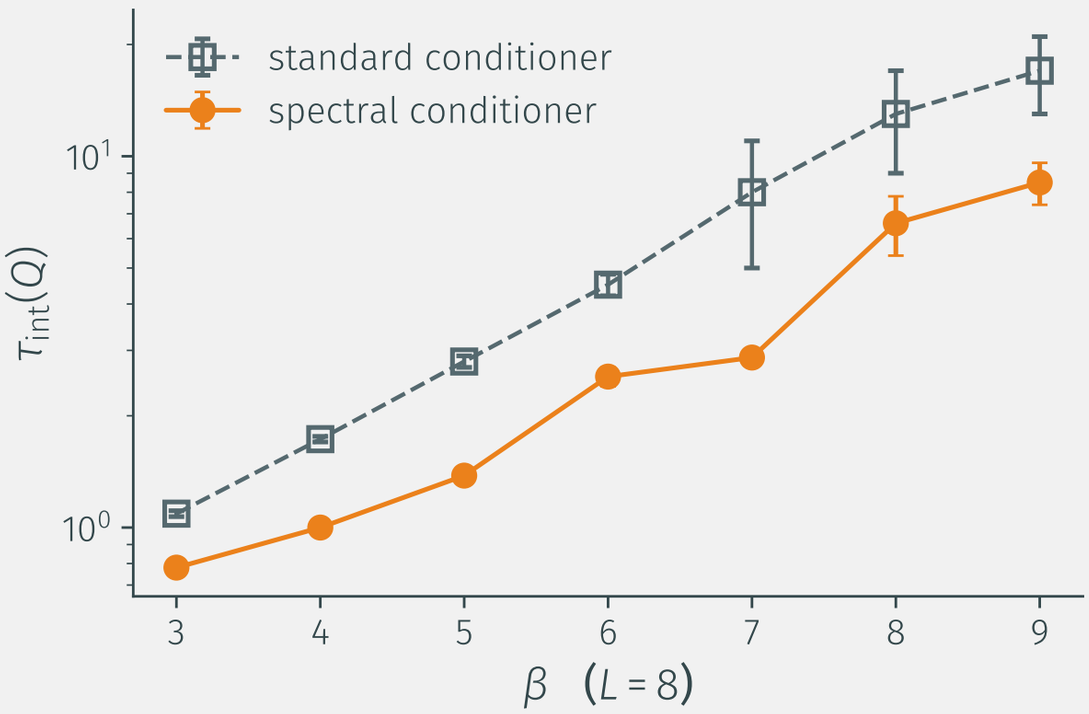


$L = 24$: χ within 1% at every β, with an error **8× smaller** from **7× fewer** samples (131k flow proposals vs 950k HMC trajectories)



$\tau_{\text{int}}(Q)$: HMC grows $\times 500$ over the range, the flow $\times 24$ → a factor **~400** at $\beta = 7$

A physics-informed spectral conditioner



At large β the information moves to the **infrared**: let the conditioner see a low-pass copy of the field

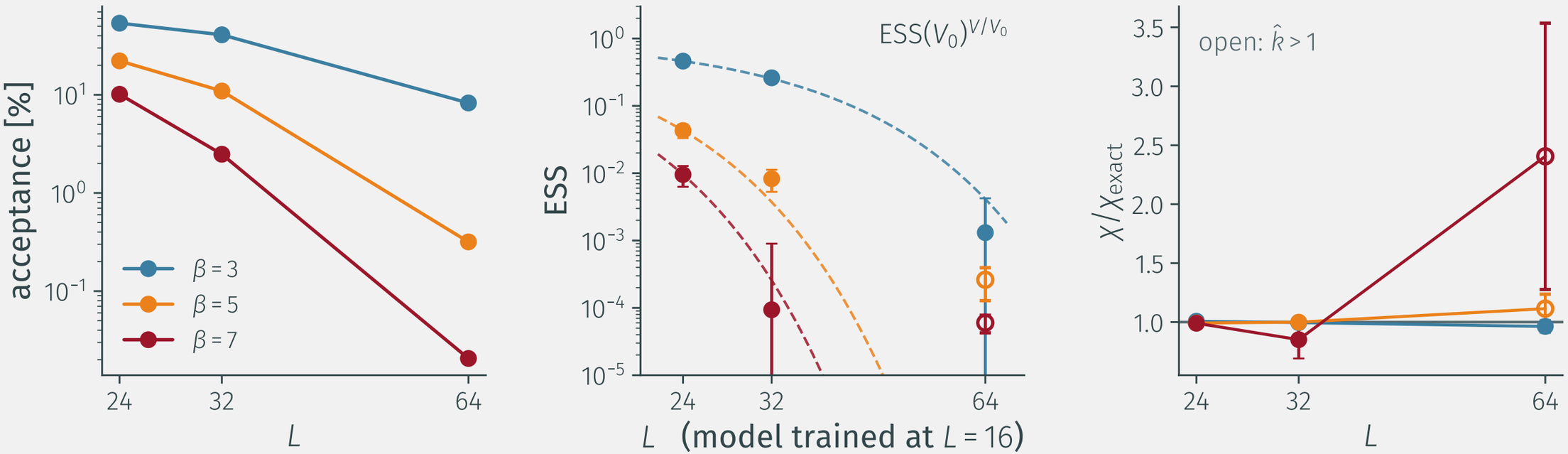
$$F_{\mathrm{LP}}(k) = \frac{1}{1 + 2^{\,k - |k_c|}}, \qquad k_c \text{ trainable}$$

The measure is untouched

The filter acts on a **copy** fed only to the conditioner: $\log\det J$ is unchanged.

$\tau_{\mathrm{int}}(Q)$ **halved**, uniformly in β

The Problem:



One model, evaluated at $L = 24 \to 64$: acceptance $10\% \to 0.02\%$ at $\beta = 7$, χ off by $\times 2.4$. Freezing reappears **inside** the flow.

$\mathrm{ESS}(V) = \mathrm{ESS}(V_0)^{V/V_0}$ [13]: V/V_0 independent regions, a small imperfection in each, **compounded**.

No single map can be trained across the whole distance on a physical lattice $\Rightarrow$ take **many small steps**



PART II · ALGORITHMIC CHALLENGES

Optimal annealing paths through distribution space

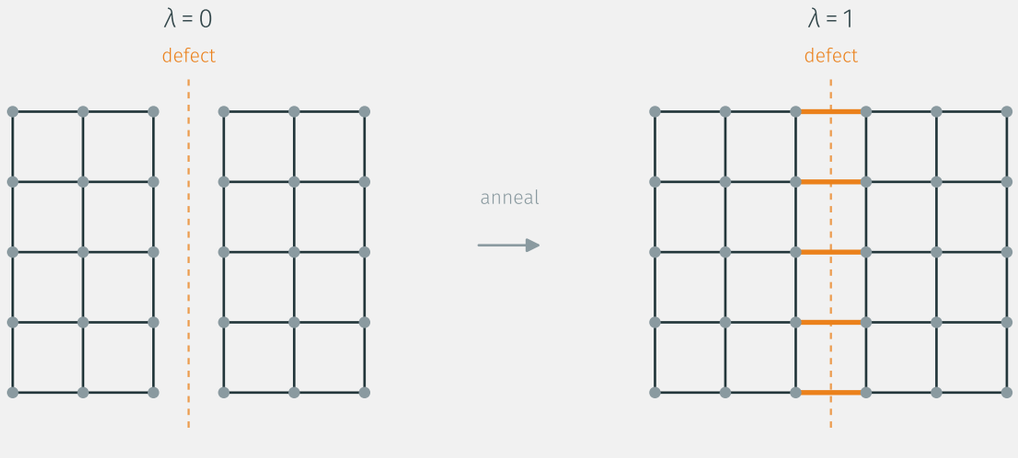
If we have to take many small steps, where should they go?

Fisher information geometry · learned coupling fields · CP^{N-1} · arxiv:2610.XXXX

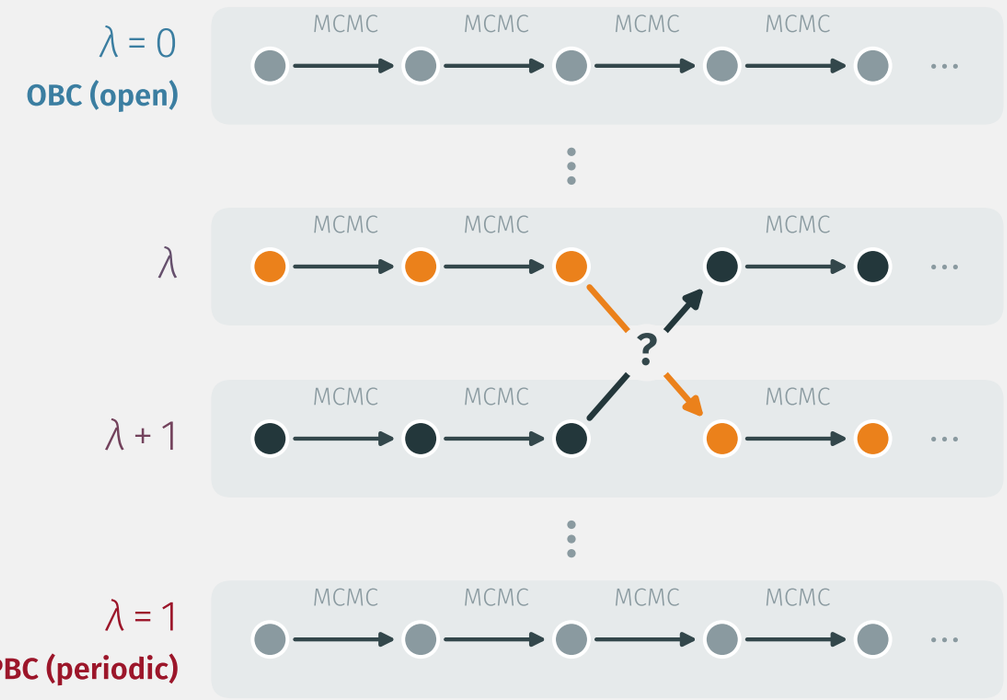
with E. Cellini, G. Kanwar, S. Widyanto

Many small steps: tempering on boundary conditions

$S_\lambda = (1-\lambda)\, S_{\text{open}} + \lambda\, S_{\text{periodic}}$, traversed **in sequence** (stochastic NFs [14]) or **in parallel** (PT)



A **defect** of links with coupling $\lambda\,\beta$:
open at $\lambda = 0$ (Q free), periodic at $\lambda = 1$ [15]



Swap neighbours with $A = \min\left(1, \dfrac{p_{\lambda_r}(x_{r+1})\, p_{\lambda_{r+1}}(x_r)}{p_{\lambda_r}(x_r)\, p_{\lambda_{r+1}}(x_{r+1})}\right)$;
measure on the periodic replica only: **exact**, and already in production in full 2+1 QCD [16]

The free parameter nobody optimises: the **schedule** $0 = \lambda_0 < \lambda_1 < \cdots < \lambda_N = 1$

Information geometry fixes the schedule

The **Fisher metric** on the family, measurable as a variance:

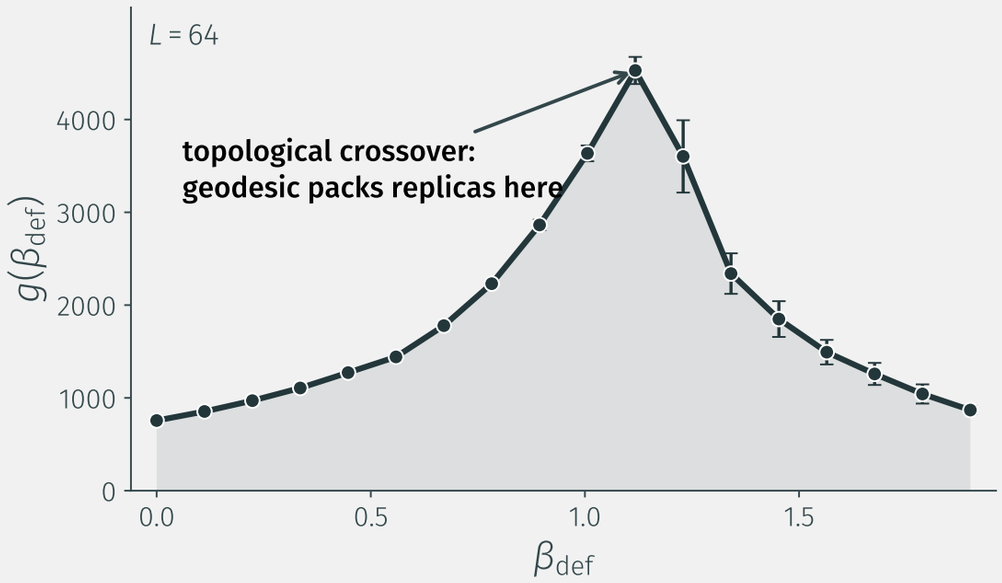
$$g(\lambda) = \mathbb{E}_{p_\lambda}\!\left[(\partial_\lambda \log p_\lambda)^2\right] = \mathbf{Var}_{p_\lambda}\!\left[\partial_\lambda S_\lambda\right]$$

$D_{\mathrm{KL}}(p_\lambda \| p_{\lambda+d\lambda}) \simeq \tfrac{1}{2} g\, d\lambda^2$: large g = hard to bridge.

Thermodynamic length [17] and dissipation

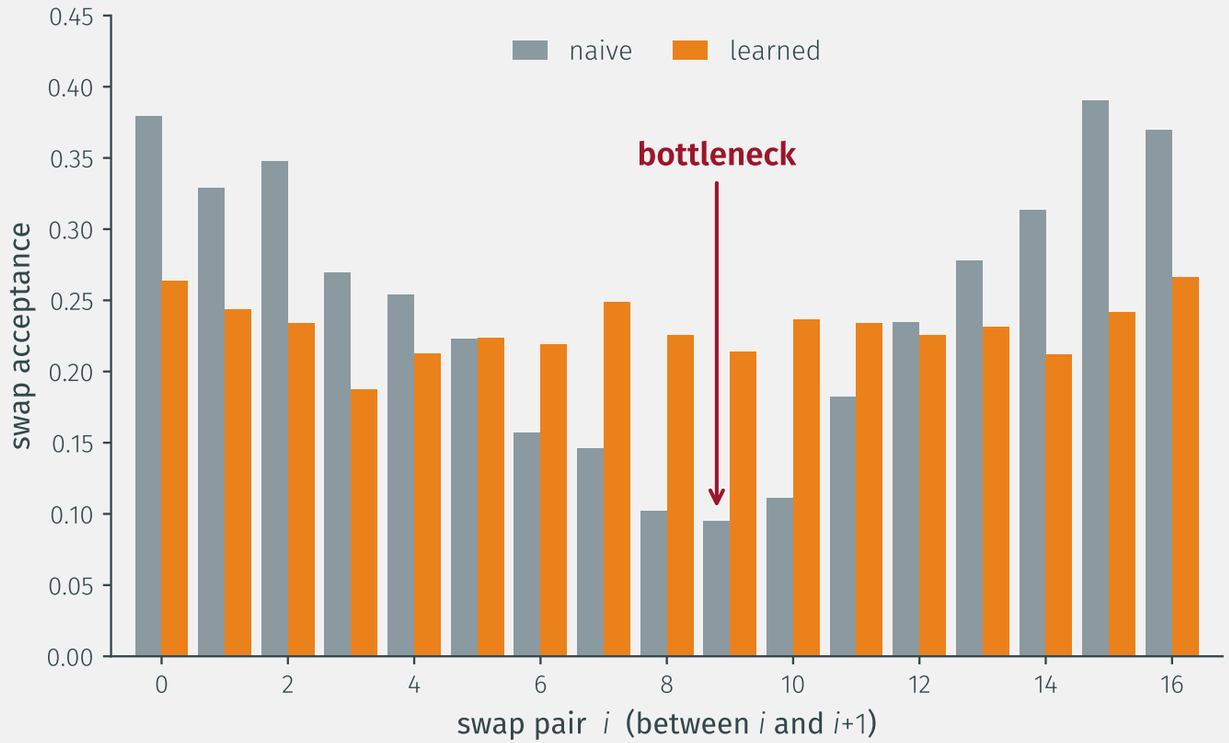
$$\Lambda = \int_0^1 \sqrt{g}\, d\lambda = \sum_k \Delta\ell_k, \qquad W_{\mathrm{diss}} \propto \sum_k \Delta\ell_k^2$$

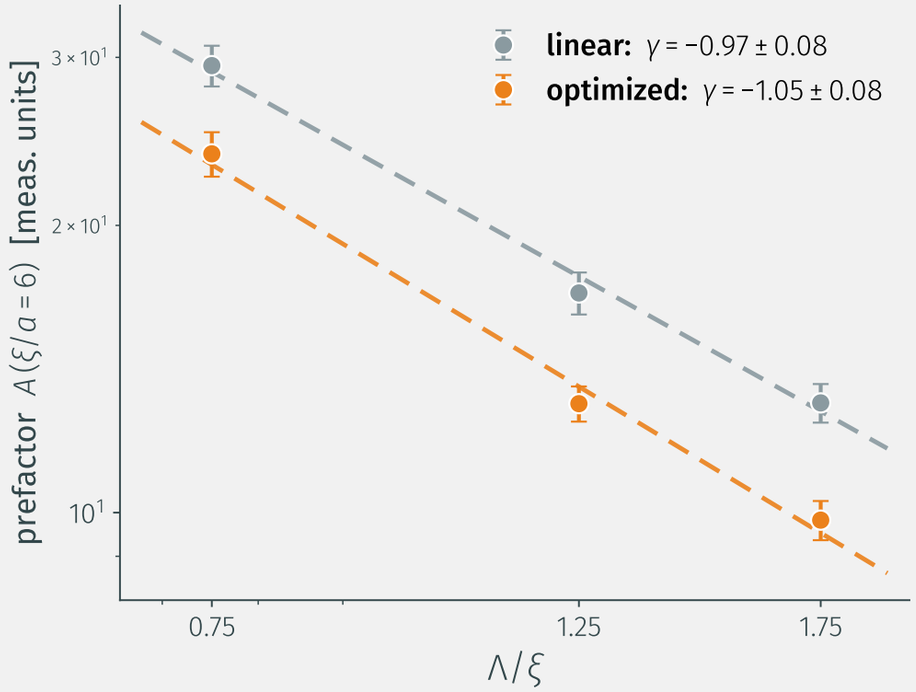
geodesic: $\Delta\ell_k = \Lambda/N \iff \dfrac{d\lambda}{dt} \propto \dfrac{1}{\sqrt{g(\lambda)}}$



g peaks at the **topological crossover**: that is where the replicas must go.

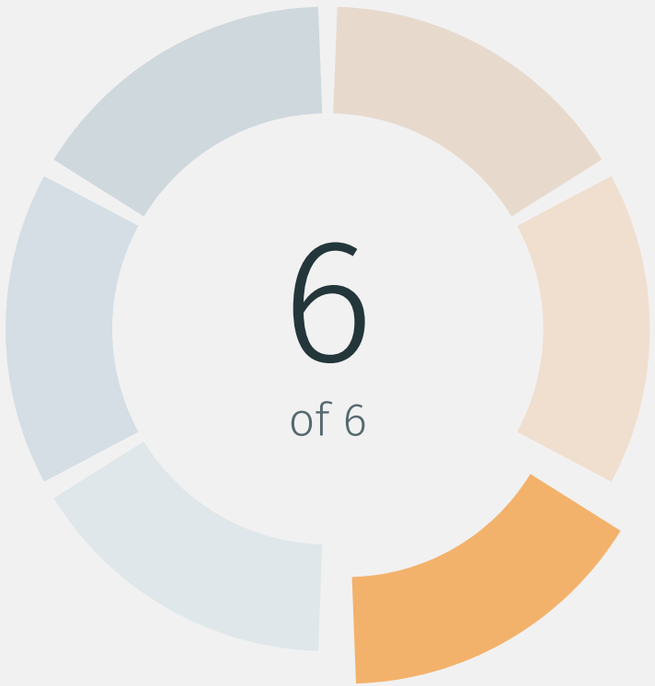
The bottleneck in the ladder disappears





- uniform ladder: swap acceptance collapses to **9.5 %** at the crossover
- optimised schedule: every pair near **22 %**, replicas round-trip;
- CP^{20}, $L = 150$: the periodic replica visits $Q = -8 \cdots + 8$

$\tau_{\text{int}}(Q^2) \sim A\,(\xi/a)^z$: a power law,
and **–20 %** on A, at no extra cost



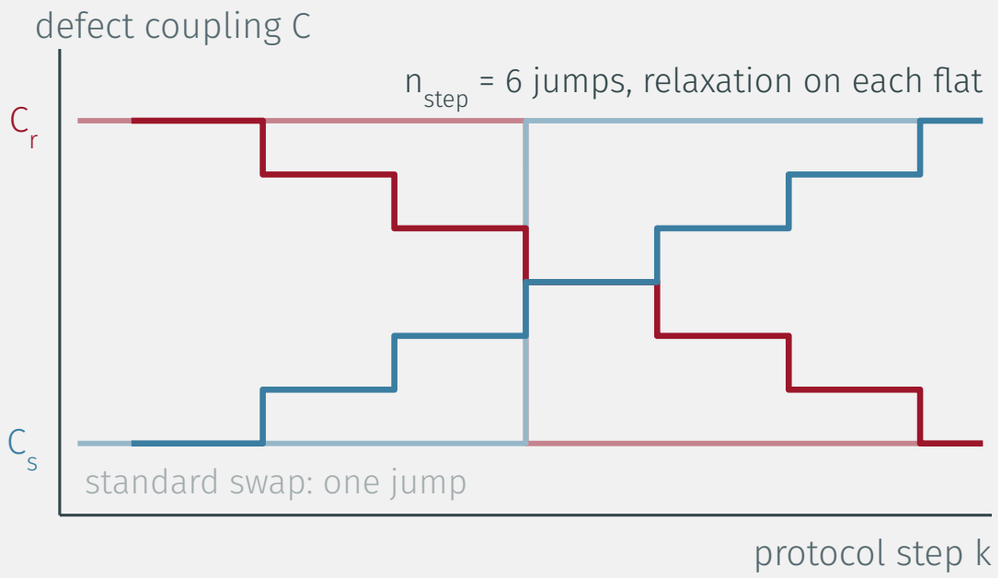
Out-of-equilibrium replica exchange

> *We chose where the replicas sit.*
> *Can the swap itself be made better?*

O-REX · exact for any protocol · CP^{20} · no neural network (yet)

arxiv:2610.XXXX

O-REX: drive the couplings past each other



n_{step} jumps with local relaxation in between [18]. The action is **linear** in the defect coupling, $S = S_{\mathrm{bulk}} + C\,E_{\mathrm{def}}$, so

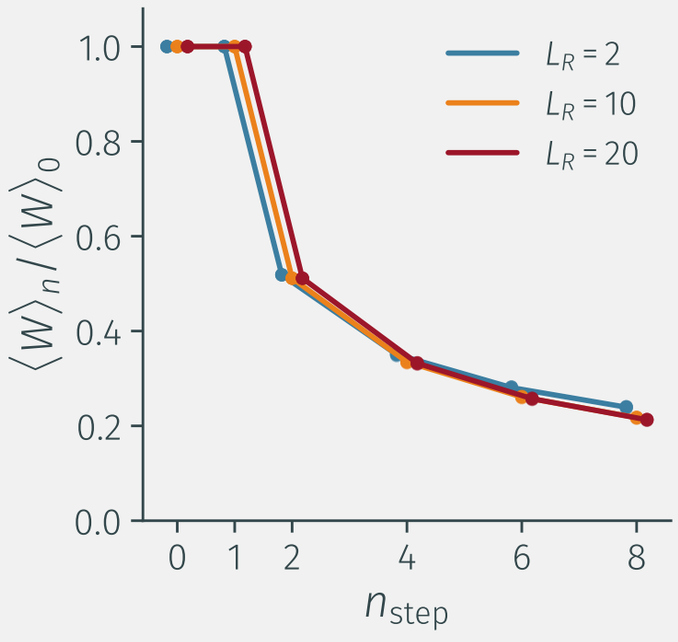
$$W = \sum_k \left[\Delta C_r^{(k)} E_{\mathrm{def}}(x_r^{(k)}) + \Delta C_s^{(k)} E_{\mathrm{def}}(x_s^{(k)})\right]$$

accept with $\min(1, e^{-W})$ and $n_{\mathrm{step}} = 0$ is the standard swap.

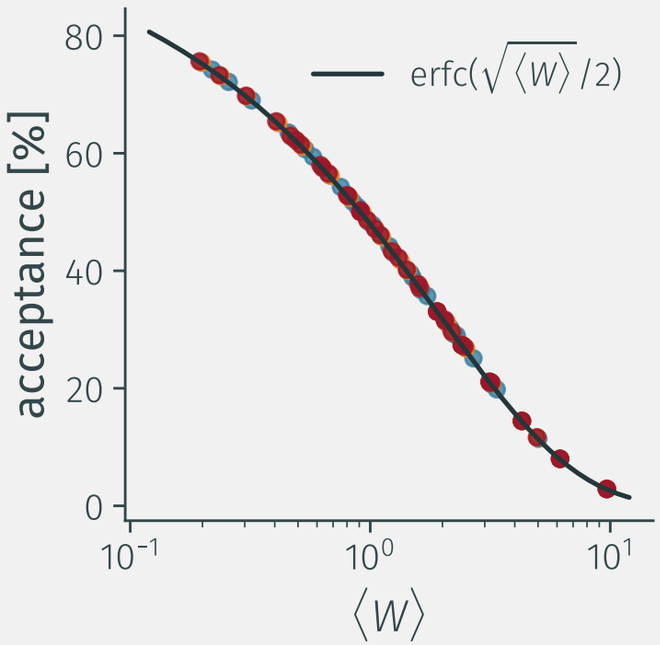
Exact for any n_{step}
Same ensembles at both ends: Jarzynski [19] gives $\langle e^{-W}\rangle = 1$, and $\langle W\rangle$ is the dissipation, i.e. a KL divergence: the flows' loss.

Made it cheap
In theory we should relax the full lattice between jumps; in practice we can relax only a patch of half-size L_R around the defect; the figure of merit is $\tau_{\mathrm{int}}(Q^2) \times \mathrm{cost}$.

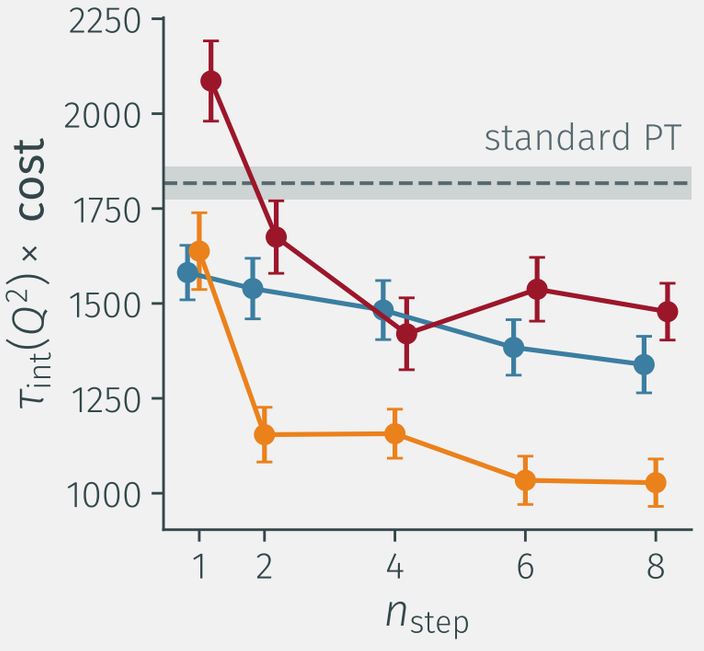
O-REX in CP^{20}: a 1.77× gain at equal cost



Dissipation cut **5×** at $n_{\mathrm{step}} = 8$, independent of N_R: $\langle W\rangle_8 / \langle W\rangle_0 = 0.217$



All 126 runs on the **parameter-free** law $\mathbf{erfc}(\sqrt{\langle W\rangle}/2)$, within 0.6 points



$\mathbf{1817 \to 1028}$: gain **1.77(12)** at $L_R = 10$; the best patch is $\sim 2\xi$

To Sum up

TOPOLOGICAL OBSERVABLES

Topological rate at finite p
first lattice determination at $p \neq 0$: $\Gamma_{\text{top}} \propto p$ up to $p/T \simeq 3\pi$, two orderings agree

b_2 across the crossover
first determination in $N_f = 2+1$: from ChPT at 130 MeV to DIGA at 200 MeV (N. Francesconi's MSc)

Lines of constant physics
step scaling: **per-mille** agreement with BMW; fermion flow: masses to **0.3 %** (WIP)

ALGORITHMIC CHALLENGES

Normalizing flows
$\tau_{\text{int}}(Q)$ **~400×** below HMC at $\beta = 7$; spectral conditioner another **2×**; ESS wall in V

Optimal annealing paths
Fisher geodesic **learned from samples**: bottleneck removed, **−20 %** on the prefactor

Out-of-equilibrium exchange
exact, controlled by the same thermodynamic length: **1.77(12)×** at equal cost

Third year plan

$\Gamma_{\text{top}}(p)$ in full QCD 1

- $N_f = 2+1$, physical masses: does $\Gamma \propto p$ survive?
- one temperature first then a scan towards **500 MeV**

The LCP, closed 2

- third lattice spacing: the **closure test** of the fermion flow
- mass step-scaling function σ_z
- Producing $2+1+1$ ensembles for the high-T plan

NEO-REX 3

- trainable gauge-equivariant (stout-like) maps between the O-REX steps
- train by minimising $\langle W \rangle$; patch $\sim \xi$, **not** V: no ESS wall
- scaling to the continuum, then 4D

enhanced sampler → ensembles on the $N_f = 2+1+1$ LCP → χ, b_2, $\Gamma_{\text{top}}(p)$ at high T

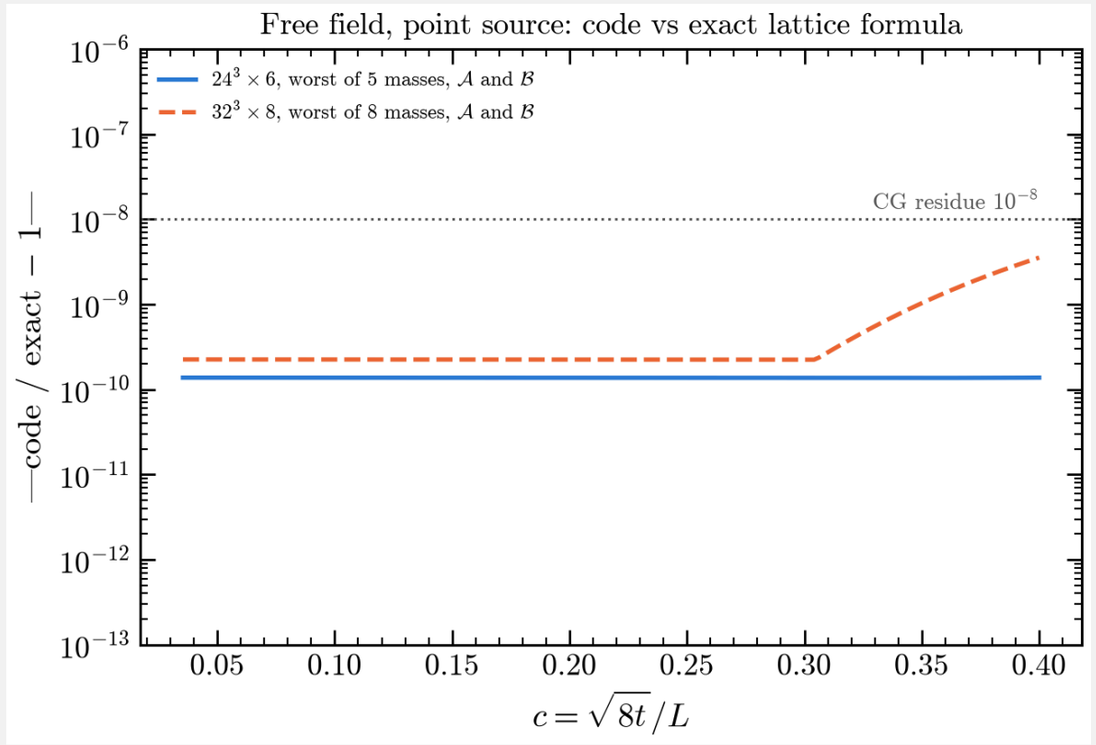
Thank you for your attention!

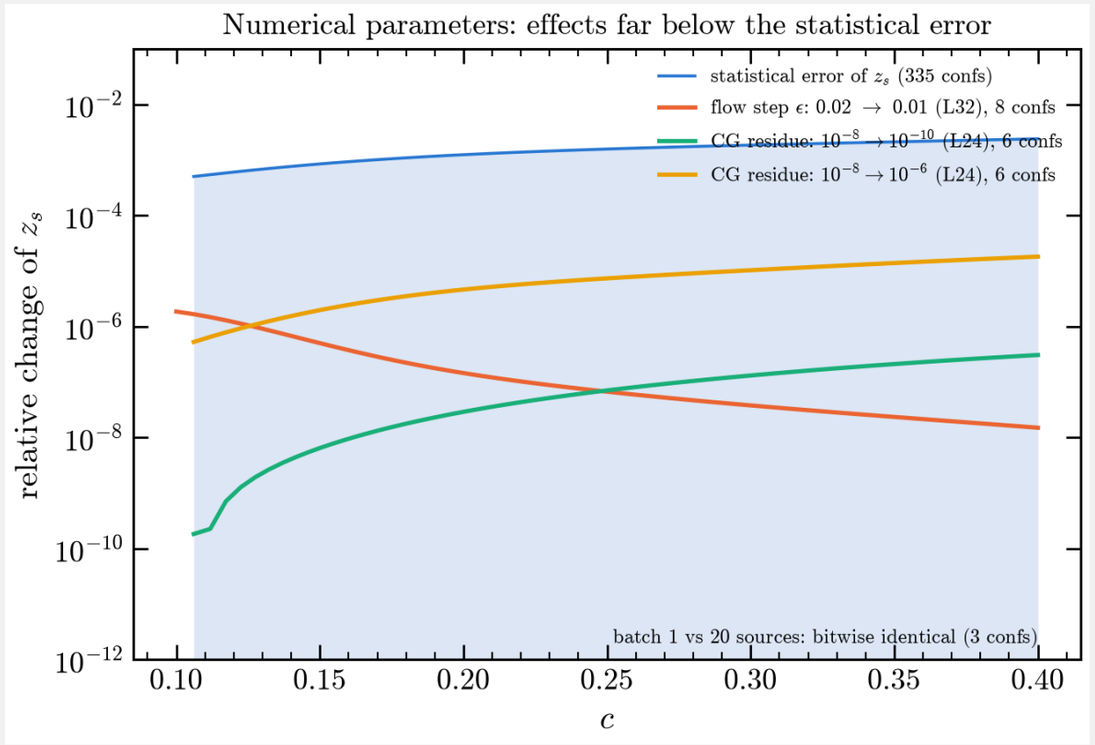
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Backup · code correctness

WORK IN PROGRESS

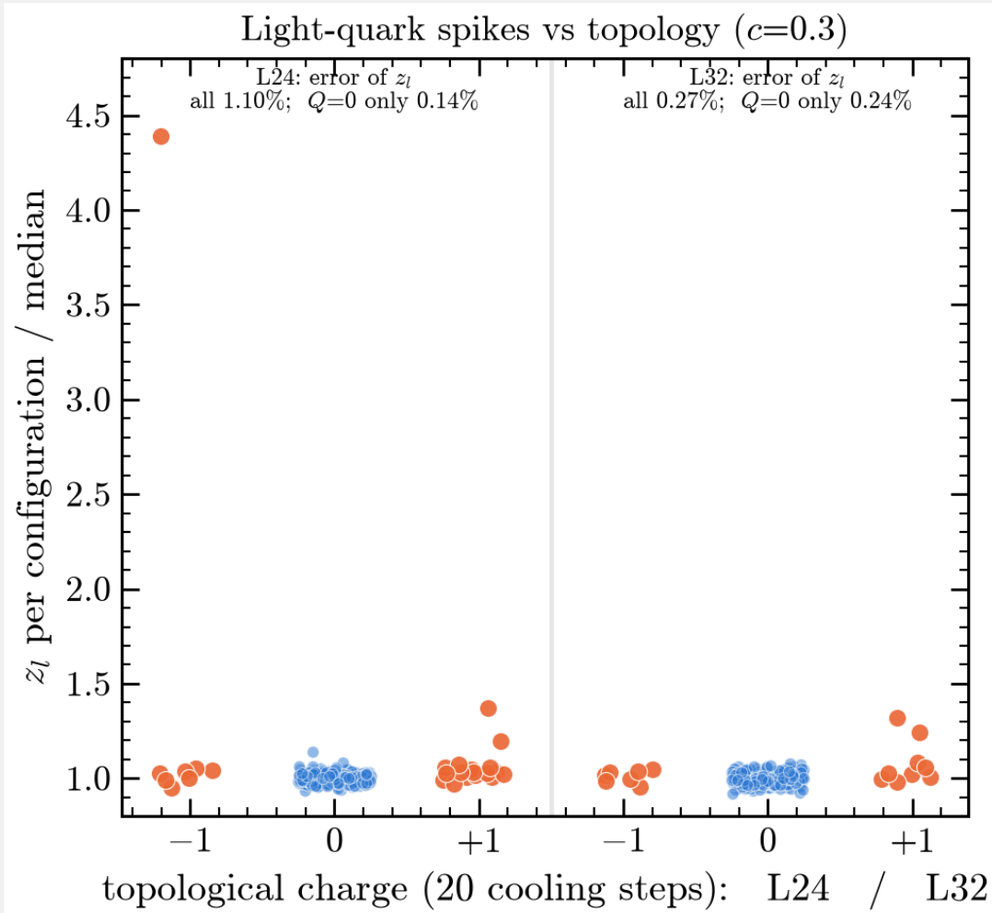


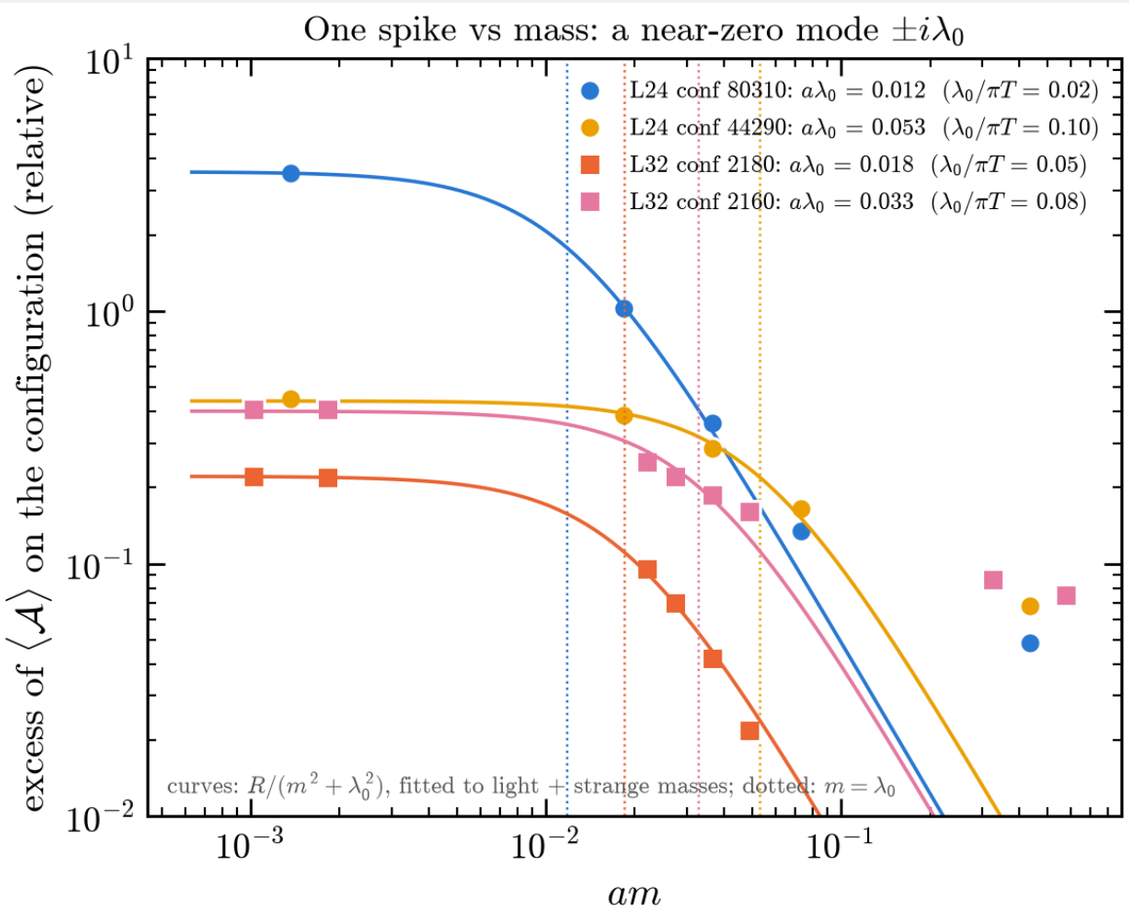


Free field: exact finite-box formula to $\mathbf{10^{-10}}$ ($\mathbf{24^3}$) and $\mathbf{4 \cdot 10^{-9}}$ ($\mathbf{32^3}$). Flow step $\mathbf{0.02 \to 0.01}$: $\mathbf{\leq 2 \cdot 10^{-6}}$; CG residue $\mathbf{10^{-8} \to 10^{-10}}$: $\mathbf{\leq 3 \cdot 10^{-7}}$; batching: bitwise identical.

Backup · the noisy light quark

WORK IN PROGRESS

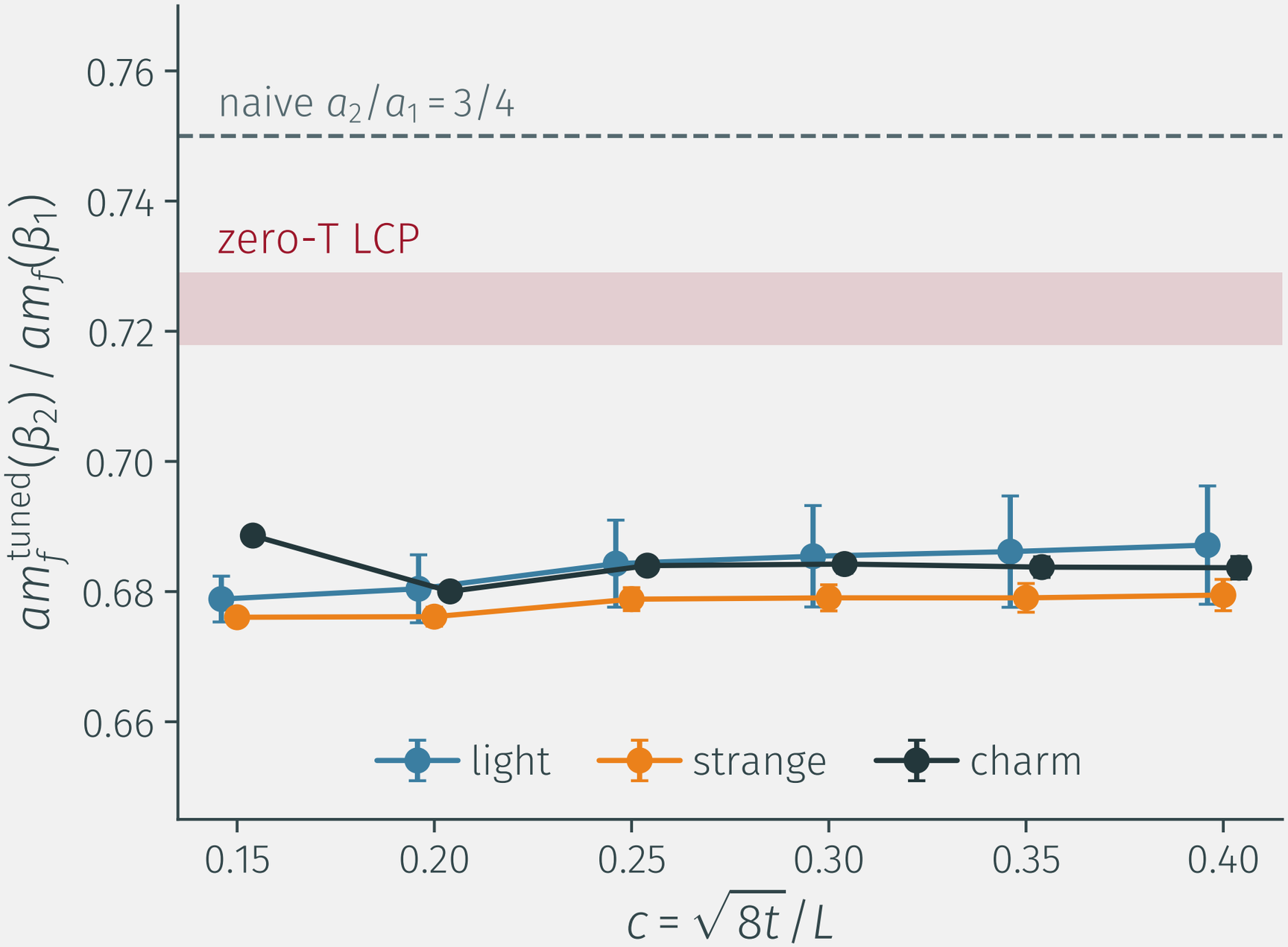




Spikes are $|Q| = 1$ configurations with one near-zero Dirac mode ($a\lambda_0 = 0.012$–0.053, fitted from the mass dependence of z). In the $Q = 0$ sector the light error drops from 1.10% to 0.14% on $24^3 \times 6$. Tune with strange: one factor for all flavours.

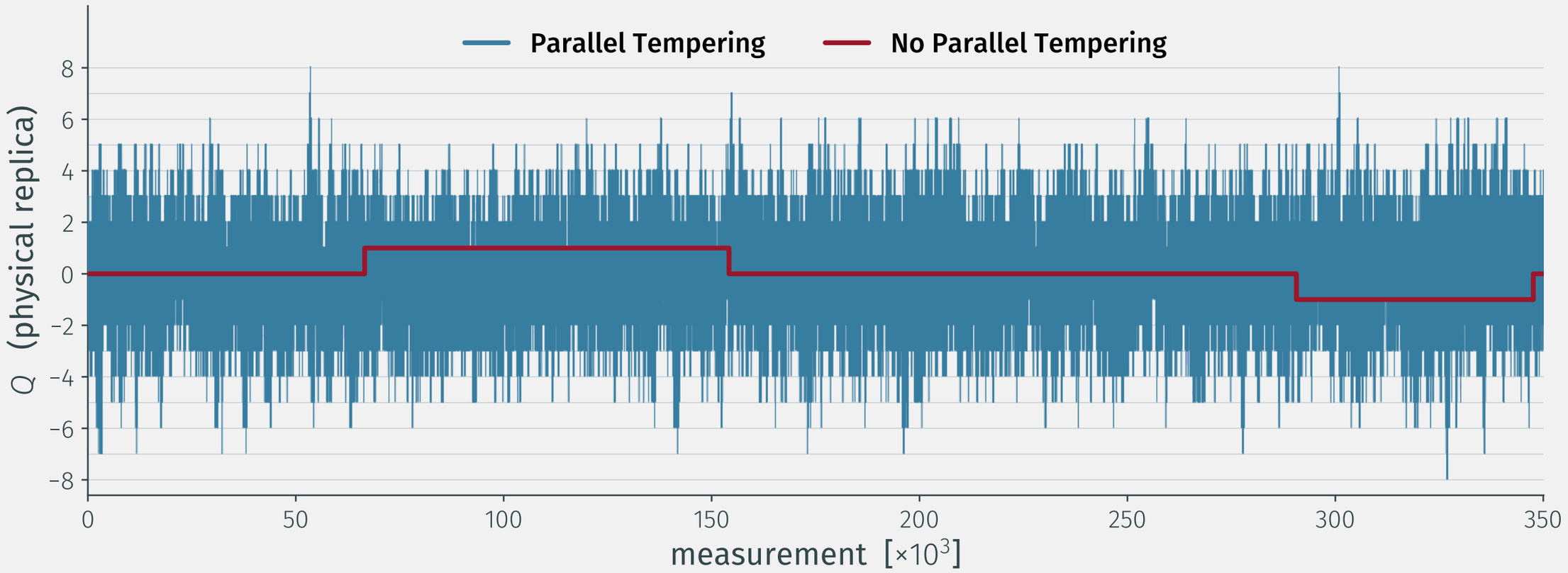
Backup · tuned ratio vs c

WORK IN PROGRESS



The tuned ratio $am(\beta_2)/am(\beta_1)$ is flat in $c \in [0.15, 0.40]$ and the same for l, s, c; the zero-T LCP sits 6–7 % higher, the naive $3/4$

Backup · topology is not frozen



CP^{20}, $L = 150$, $N_R = 12$, optimised schedule, ~350k measurements: the periodic replica visits $Q = -8 \cdots + 8$.